

COVER SHEET FOR PROPOSAL TO THE NATIONAL SCIENCE FOUNDATION

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TITLE OF PROPOSED PROJECT Multigraded Homological Algebra and Geometry					SHOW LETTER OF INTENT ID IF APPLICABLE
REQUESTED AMOUNT \$ 357,582	PROPOSED DURATION (1-60 MONTHS) 24 months	REQUESTED STARTING DATE 05/01/2026		SHOW RELATED PRELIMINARY PROPOSAL NO. IF APPLICABLE	
THIS PROPOSAL INCLUDES ANY OF THE ITEMS LISTED BELOW					
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CERTIFICATION PAGE

Certification for Authorized Organizational Representative (or Equivalent)

By electronically signing and submitting this proposal, the Authorized Organizational Representative (AOR) is: (1) certifying that statements made herein are true and complete to the best of the individual's knowledge; and (2) agreeing to accept the obligation to comply with NSF award terms and conditions if an award is made as a result of this proposal. Further, the proposer is hereby providing certifications regarding conflict of interest, flood hazard insurance, responsible and ethical conduct of research, organizational support, and safe and inclusive working environments for off-campus or off-site research, as set forth in the NSF Proposal & Award Policies & Procedures Guide (PAPPG). Willful provision of false information in this application and its supporting documents or in reports required under an ensuing award is a criminal offense (U.S. Code, Title 18, Section §1001).

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(This Certification applies to proposals submitted prior to July 31, 2023, and is not applicable to proposals for conferences, symposia, and workshops.)

By electronically signing the Certification Pages, the Authorized Organizational Representative is certifying that, in accordance with the NSF Proposal & Award Policies & Procedures Guide, Chapter IX.B., the institution has a plan in place to provide appropriate training and oversight in the responsible and ethical conduct of research to undergraduates, graduate students and postdoctoral researchers who will be supported by NSF to conduct research. The AOR shall require that the language of this certification be included in any award documents for all subawards at all tiers.

Certification Regarding Responsible and Ethical Conduct of Research (RECR)

(This Certification applies to proposals submitted on or after July 31, 2023, and is not applicable to proposals for conferences, symposia, and workshops.)

By electronically signing the Certification Pages, the Authorized Organizational Representative is certifying that, in accordance with the NSF Proposal & Award Policies and Procedures Guide, Chapter IX.B., the institution has a plan in place to provide appropriate training and oversight in the responsible and ethical conduct of research to undergraduate students, graduate students, postdoctoral researchers, faculty, and other senior/key personnel who will be supported by NSF to conduct research. As required by Section 7009 of the America Creating Opportunities to Meaningfully Promote Excellence in Technology, Education, and Science (COMPETES) Act (42 U.S.C 18620-1), as amended, the training addresses mentor training and mentorship.

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Certification Regarding Organizational Support

By electronically signing the Certification Pages, the Authorized Organizational Representative (or equivalent) is certifying that there is organizational support for the proposal as required by Section 526 of the America COMPETES Reauthorization Act of 2010. This support extends to the portion of the proposal developed to satisfy the Broader Impacts Review Criterion as well as the Intellectual Merit Review Criterion, and any additional review criteria specified in the solicitation. Organizational support will be made available, as described in the proposal, in order to address the broader impacts and intellectual merit activities to be undertaken.

Certification Regarding Dual Use Research of Concern

By electronically signing the certification pages, the Authorized Organizational Representative is certifying that the organization will be or is in compliance with all aspects of the United States Government Policy for Institutional Oversight of Life Sciences Dual Use Research of Concern.

Certification Requirement Specified in the William M. (Mac) Thornberry National Defense Authorization Act for Fiscal Year 2021, Section 223(a)(1) (42 USC 6605(a)(1))

By electronically signing the Certification Pages, the Authorized Organizational Representative is certifying that each individual employed by the organization and identified on the proposal as senior/key personnel has been made aware of the certification requirements identified in the William M. (Mac) Thornberry National Defense Authorization Act for Fiscal Year 2021, Section 223(a)(1) (42 U.S.C § 6605(a)(1)).

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(This certification applies only to proposals in which data/information/samples are being collected off-campus or off-site, such as fieldwork and research activities on vessels and aircraft.) By electronically signing the Certification Pages, the Authorized Organizational Representative is certifying that, in accordance with the NSF Proposal & Award Policies and Procedures Guide, Chapter II.E.9, the organization has a plan in place for this proposal regarding safe and inclusive working environments.

Certification Regarding Malign Foreign Talent Recruitment Programs

By electronically signing the Certification Pages, the Authorized Organizational Representative is certifying that, in accordance with Section 10632 of the CHIPS and Science Act of 2022 (42 U.S.C. 19232), all senior/key personnel associated with the proposal have been made aware of and have complied with their responsibility under that section to certify that they are not a party to a malign foreign talent recruitment program.

AUTHORIZED ORGANIZATIONAL REPRESENTATIVE		SIGNATURE	DATE
NAME			
TELEPHONE NUMBER	EMAIL ADDRESS	FAX NUMBER	

Project Summary: Multigraded Homological Algebra & Geometry

Overview.

This proposal advances the study of multigraded homological algebra, algebraic geometry, and combinatorics. It has three integrated projects: i) establishing connections between minimal graded free resolutions and geometry on stacky curves and toric stacks, ii) clarifying the connection between multigraded Castelnuovo–Mumford regularity and free and virtual resolutions on toric varieties, and providing bounds and algorithms for multigraded regularity, and iii) building a uniform framework to study homological invariants across all toric varieties with the same Cox ring. The PI expects the outcomes of this proposal to include new theorems linking homological algebra and toric geometry, developing open-source software, and targeted efforts to expand the mathematical community.

Intellectual Merit.

The first project seeks to generalize classical results about the syzygies of algebraic varieties, e.g., Green’s N_p -theorem for curves of high degree, Green’s conjecture concerning the syzygies of canonical curves, Ein and Lazarsfeld’s work on asymptotic syzygies, to stacky curves and toric stacks. As part of this work, the PI proposes a number of concrete questions developing a suitable theory of kernel bundles and Koszul cohomology adapted to the multigraded setting of toric stacks and stacky curves, studying non-standard graded analogs of Koszulness and their implications for the multigraded Betti numbers of stacky curves. The PI plans to develop software for computing with toric stacks and expand the database of syzygy computations the PI maintains.

The PI’s second project aims to further understanding of multigraded Castelnuovo–Mumford regularity, and other measures of algebraic and geometric complexity, in the multigraded and toric settings. As part of this, the PI seeks to generalize her recent work showing the precise relationship between free resolutions, virtual resolutions, and multigraded regularity on products of projective spaces to arbitrary toric varieties. The PI proposes several directions to develop meaningful bounds on multigraded regularity and to study its asymptotic behavior. As a consequence of this work, the PI hopes to develop algorithms for effectively computing multigraded regularity.

The third project expands upon a developing theme that many homological properties of toric varieties only depend on the underlying graded ring. The main direction of this project is developing uniform homological algebra tools (multigraded regularity and virtual resolutions) in the D_{Cox} category. Using these tools, the PI hopes to prove: i) uniform vanishing and regularity statements that hold for all toric varieties with a fixed Cox ring, ii) descriptions of controlled changes in multigraded Betti numbers under variation of GIT stability, and iii) relate the local cohomology of modules after saturating by different monomial ideals corresponding to sheaves on different quotients. This project is closely connected to the first two projects.

Broader Impacts.

The PI will continue her work to broaden the mathematical community and develop the next generation of mathematicians. This includes continued mentoring of two undergraduates and three graduate students, organizing the *Unspoken Rules* seminar, providing professional development to graduate students facing additional career challenges, and leading a project at a summer undergraduate research program in 2026. The PI has served in a number of roles seeking to support the growth and promotion of the mathematical community. This has included serving on committees for the AMS and SLMath, and being the inaugural president of *Spectra*. The PI will continue serving on AMS committees, and plans to re-engage in organizing for *Spectra*. The PI has organized several conferences: *AGNES* (2024), *GEMS of Commutative Algebra* (2019, 2023), *Spec($\overline{\mathbb{Q}}$)* (2022), *WAGS* (2021), *GEMS of Combinatorics* (2023), *Geometry & Arithmetic of Surfaces* (2019), and *M2@UW* (2018). The PI plans to organize: *BATMOBILE* in Fall 2025, *GEMS of Commutative Algebra* in Winter 2027, and is preparing a BIRS proposal for a conference on syzygies and geometry.

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Project Description: Multigraded Homological Algebra and Geometry

This proposal involves research in algebraic geometry and commutative algebra with several connections to computation and combinatorics, as well as a wide array of broader impacts.

- § 1 Results of Prior NSF Support.
- § 2 The Geometry of Multigraded Syzygies on Stacks
- § 3 Castelnuovo–Mumford Regularity: Resolutions, Bounds, and Asymptotic Behavior
- § 4 Uniform Homological Algebra on Toric Varieties
- § 5 Broader Impacts.

1. Results of Prior NSF Support

During 2020–2022 the PI was supported by an NSF Postdoctoral Research Fellowship (NSF Grant No. MSPRF DMS-2002239, \$150,000) titled *Asymptotic Syzygies in Algebraic Geometry*. In this period the PI made major contributions to extending our understanding of homological algebra on toric varieties (Section 1.2) and the cohomology of moduli spaces (Section 1.4). These works resulted in 6 papers being posted to the arXiv [23, 28, 27, 16, 15, 4].

During 2015–2020 the PI was supported by an NSF Graduate Research Fellowship (NSF Grant No. DGE-1256259, \$150,000) title *Syzygies in Algebraic Geometry*. In this period the PI made substantial contributions to commutative algebra and algebraic geometry, including furthering our understanding of syzygies on higher dimensional varieties. This resulted in the PI posting 8 new papers to the arXiv [1, 14, 24, 29, 25, 26, 22, 20].

Of the 14 new papers the PI posted to the arXiv during these periods of support 10 have been accepted for publication, including in: *Algebra & Number Theory* [24], *Geometry & Topology* [15], *Journal of Algebra* [23], and *Experimental Mathematics* [25]. The PI contributed to 4 open-source software packages, one public-facing database, and two articles for the *Notices of the AMS* [11, 21]. A detailed summary of these results and their intellectual merits continues in the following sections.

1.1 Asymptotic Syzygies Broadly, asymptotic syzygies is the study of the graded Betti numbers (i.e. the syzygies) of a projective variety as the positivity of the embedding grows. For a smooth projective variety $X \subset \mathbb{P}^{n_d}$ embedded by a very ample line bundle L_d ; following [49], set,

$$\rho_q(X, L_d) := \frac{\#\{p \in \mathbb{N} \mid \beta_{p,p+q}(X, L_d) \neq 0\}}{n_d},$$

which is the percentage of non-zero syzygies appearing in row q of the Betti table [47, Theorem 1.1]. The asymptotic perspective asks how $\rho_q(X; L_d)$ behaves along the sequence of line bundles $(L_d)_{d \in \mathbb{N}}$. In this language, Green’s work on the syzygies of high degree curves can be phrased as follows.

Theorem 1.1. [61] *Let $X \subset \mathbb{P}^n$ be a smooth projective curve. If $(L_d)_{d \in \mathbb{N}}$ is a sequence of very ample line bundles on X such that $\deg L_d = d$ then $\lim_{d \rightarrow \infty} \rho_2(X; L_d) = 0$.*

Put differently, asymptotically the syzygies of curves are as simple as possible, occurring in the lowest possible degree. Ein and Lazarsfeld established the opposite asymptotic behavior for higher dimensional varieties under the analogous positivity condition.

Theorem 1.2. [46, Theorem C] *Let $X \subset \mathbb{P}^n$ be a smooth projective variety, $\dim X \geq 2$, and fix an index $1 \leq q \leq \dim X$. If $(L_d)_{d \in \mathbb{N}}$ is a sequence of very ample line bundles such that $L_{d+1} - L_d$ is constant and ample then $\lim_{d \rightarrow \infty} \rho_q(X; L_d) = 1$.*

My work weakens the ample-growth hypothesis to semi-ample growth. Recall that a line bundle L is *semi-ample* if $|kL|$ is base point free for $k \gg 0$ (e.g. $\mathcal{O}(1, 0)$ on $\mathbb{P}^n \times \mathbb{P}^m$). I established the following asymptotic nonvanishing result for $PP^n \times \mathbb{P}^m$ embedded by $\mathcal{O}(d_1, d_2)$.

Theorem 1.3. [22, Corollary B] *Let $X = \mathbb{P}^n \times \mathbb{P}^m$ and fix an index $1 \leq q \leq n + m$. There exist constants $C_{i,j}$ and $D_{i,j}$ such that*

$$\rho_q(X; \mathcal{O}(d_1, d_2)) \geq 1 - \sum_{\substack{i+j=q \\ i \leq n, j \leq m}} \left(\frac{C_{i,j}}{d_1^i d_2^j} + \frac{D_{i,j}}{d_1^{n-i} d_2^{m-j}} \right) - O\left(\frac{\text{lower ord.}}{\text{terms}}\right).$$

If both $d_1 \rightarrow \infty$ and $d_2 \rightarrow \infty$ then $\rho_q(\mathbb{P}^n \times \mathbb{P}^m; \mathcal{O}(d_1, d_2)) \rightarrow 1$, recovering the results of Ein and Lazarsfeld for $\mathbb{P}^n \times \mathbb{P}^m$. However, if d_1 is fixed and $d_2 \rightarrow \infty$ (i.e. semi-ample growth) my results bound the asymptotic percentage of non-zero syzygies away from zero. In fact I conjecture that, unlike in previously studied cases, in the semi-ample setting $\rho_q(\mathbb{P}^n \times \mathbb{P}^m; \mathcal{O}(d_1, d_2))$ does not approach 1. Proving this would require a vanishing result for asymptotic syzygies, which is open even in the ample case [46, Conjectures 7.1, 7.5].

1.2 Eisenbud-Goto on Products of Projective Space The Castelnuovo–Mumford Regularity of a projective variety $X \subset \mathbb{P}^r$ is a measure of the complexity of X given in terms of the vanishing of certain cohomology groups of X . Roughly speaking one should think about Castelnuovo–Mumford regularity as being a numerical measure of geometric complexity. Eisenbud and Goto showed that regularity is closely connected to interesting homological properties.

Theorem 1.4. [48] *Let $\mathcal{F}$ be a coherent sheaf on $\mathbb{P}^n$ and $M = \bigoplus_{e \in \mathbb{Z}} H^0(\mathbb{P}^n, \mathcal{F}(e))$ the corresponding section module. The following are equivalent: (1) M is d -regular; (2) $\beta_{p,q}(M) = 0$ for all $p \geq 0$ and $q > d + i$; (3) $M_{\geq d}$ has a linear resolution.*

A toric variety Y is a well-behaved compactification of a torus, which under minor hypotheses, can be written as a quotient $Y = [(\mathbb{C}^n \setminus \mathbb{V}(B))/(\mathbb{C}^\times)^r]$ where B is the *irrelevant ideal* of Y ; for $Y \cong \mathbb{P}^n$ the irrelevant ideal $B = \langle x_0, \dots, x_n \rangle$. Cox proved a beautiful correspondence between coherent sheaves on Y and $\mathbb{Z}^r$ -graded (B -saturated) modules over a $\mathbb{Z}^r$ -graded polynomial ring $S = \mathbb{C}[x_0, \dots, x_n]$, which is known as the Cox ring of Y [36]. This yields a rich dictionary between *multigraded* homological algebra over S and the geometry of Y . MacLagan and Smith defined multigraded Castelnuovo–Mumford regularity on toric varieties in terms of the vanishing of cohomology groups, Let Y be a smooth projective toric variety so that $\text{Pic}(Y) \cong \mathbb{Z}^r$ and the nef cone of Y is a full-dimensional cone of $\text{Pic}(Y)_{\mathbb{R}}$. For $\mathbf{v} \in \mathbb{Z}^r$ set $|\mathbf{v}| := v_1 + v_2 + \dots + v_r$.

Definition 1.5. [92, Definition 6.1] *With the set-up as in the previous paragraph if $\mathcal{F}$ is a coherent module on X then we say that $\mathcal{F}$ is $\mathbf{d}$ -regular for $\mathbf{d} \in \mathbb{Z}^r$ if and only if $H^i(Y, \mathcal{F}(\mathbf{e})) = 0$ for all $i > 0$ and all $\mathbf{e} \in L_i(\mathbf{d})$ where*

$$L_i(\mathbf{d}) := \bigcup_{\mathbf{v} \in \mathbb{Z}^r, |\mathbf{v}|=i} (\mathbf{d} - \mathbf{v} + \text{Nef}(X)).$$

We then define the multigraded regularity of $\mathcal{F}$ to be $\text{reg}(\mathcal{F}) := \{\mathbf{d} \in \mathbb{Z}^r \mid \mathcal{F} \text{ is } \mathbf{d}\text{-regular}\}$.

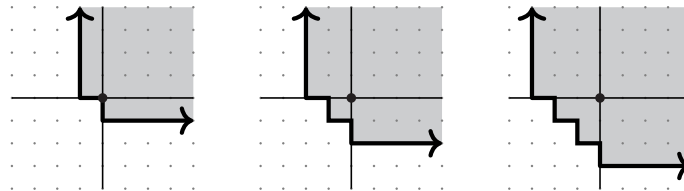


FIGURE 1. The regions $L_1(0,0)$, $L_2(0,0)$, and $L_3(0,0)$ for $Y = \mathbb{P}^n \times \mathbb{P}^m$, in which case $\text{Pic}(Y) \cong \mathbb{Z}^2$ and $\text{Nef}(Y)$ is the first quadrant.

We can define the multigraded regularity of a $\mathbb{Z}^r$ -graded module M over S in terms of analogous vanishing of local cohomology modules $H_B^I(M)$ where B is the irrelevant ideal of Y . The multigraded Castelnuovo–Mumford regularity is not a single number, but an infinite subset of $\mathbb{Z}^r$.

The obvious approaches to generalize Theorem 1.4 to a product of projective spaces fail. For example, the multigraded Betti numbers do not determine multigraded Castelnuovo–Mumford regularity [28, Example 5.1]. Despite this we show that part (3) of Theorem 1.4 can be generalized. To do so we introduce the following generalization of linear resolutions.

Definition 1.6. *A complex $F_\bullet$ of $\mathbb{Z}^r$ -graded free S -modules is **d**-quasilinear if and only if F_0 is generated in degree **d** and each twist of F_i is contained in $L_{i-1}(\mathbf{d} - 1)$.*

Theorem 1.7. [28, Theorem A] *Let M be a (saturated) finitely generated $\mathbb{Z}^r$ -graded S -module:*

$$M \text{ is } \mathbf{d}\text{-regular} \iff M_{\geq \mathbf{d}} \text{ has a } \mathbf{d}\text{-quasilinear resolution.}$$

The proof of Theorem 1.7 is based on a spectral sequence argument relating the Betti numbers of $M_{\geq \mathbf{d}}$ to the Fourier–Mukai transform of $\widetilde{M}$ with Beilinson’s resolution of the diagonal as the kernel. This result provides an effective way to compute whether a module is regular at a given multidegree $\mathbf{d} \in \mathbb{Z}^r$ and construct so called “short virtual resolutions” on products of projective spaces.

1.3 Asymptotic Multigraded Regularity of Ideals Building upon our work discussed above – and inspired by previous work on $\mathbb{P}^n$ [10, 41, 86] – I studied the asymptotic regularity of powers of ideals on arbitrary toric varieties. My collaborators and I show that the multigraded regularity of powers of ideals is bounded and translates in a predictable way. In particular, the regularity of I^t essentially translates within $\text{Nef}(Y)$ in fixed directions at a linear rate.

Theorem 1.8. [27, Theorem 4.1] *There exists a degree $\mathbf{a} \in \text{Pic}(Y)$, depending only on I , such that for each integer $t > 0$ and each pair of degrees $\mathbf{q}_1, \mathbf{q}_2 \in \text{Pic}(Y)$ satisfying $\mathbf{q}_1 \geq \deg f_i \geq \mathbf{q}_2$ for all generators f_i of I , we have*

$$t\mathbf{q}_1 + \mathbf{a} + \text{reg } S \subseteq \text{reg}(I^t) \subseteq t\mathbf{q}_2 + \text{Nef}(Y).$$

1.3.1 Syzygies via Highly Distributed Computing It is quite difficult to compute examples of syzygies. For example, until recently the syzygies of the projective plane embedded by the d -uple Veronese embedding were only known for $d \leq 5$. My co-authors and I exploited recent advances in numerical linear algebra and high-throughput high-performance computing to generate a number of new examples of Veronese syzygies. A follow-up project used similar computational approaches to compute the syzygies of $\mathbb{P}^1 \times \mathbb{P}^1$ in over 200 new examples. This data provided support for several existing conjectures, as well as led us to make a number of new conjectures [25, 23]. The data is publicly available via SyzygyData.com and a package for Macaulay2 [24, 60].

1.4 Top-Weight Cohomology of $\mathcal{A}_g$ The moduli stack of abelian varieties of dimension g , $\mathcal{A}_g$, is the smooth DM stack $[\mathbb{H}_g/\text{Sp}(2g, \mathbb{Z})]$ parameterizing isomorphism classes of principally polarized abelian varieties of dimension g . Despite extensive study, much of its geometry is unknown, for example its rational cohomology is fully known only for $g \leq 4$ [70, 65, 68, 69] and Grushevsky asked if $H^{2k+1}(\mathcal{A}_g; \mathbb{Q}) \neq 0$ for some g and k [63]. Building on [31], my co-authors and I developed methods to analyze a canonical quotient of $H^k(\mathcal{A}_g; \mathbb{Q})$ and produced new nontrivial classes.

Theorem 1.9. [15, Theorem A] *The rational cohomology $H^k(\mathcal{A}_g; \mathbb{Q}) \neq 0$ for:*

$$(g, k) = (5, 15), (5, 20), (6, 30), (7, 28), (7, 33), (7, 37), \text{ and } (7, 42).$$

This theorem provided an answer to Grushevsky. Since $\mathcal{A}_g$ is a rational classifying space for $\text{Sp}_{2g}(\mathbb{Z})$ we have $H^*(\mathcal{A}_g; \mathbb{Q}) \cong H^*(\text{Sp}_{2g}(\mathbb{Z}); \mathbb{Q})$, and thus, Theorem 1.9 gives new non-vanishing results for $\text{Sp}_{2g}(\mathbb{Z})$. Our approach uses Deligne’s mixed Hodge theory and tropical geometry to

compute the top-weight cohomology of $\mathcal{A}_g$ for $g \leq 7$. Subsequent work of the PI generalized this to tropicalizing Shimura varieties [4], highlighted in a 2025 SRI in Algebraic Geometry plenary.

1.5 Broader Impacts from Prior NSF Support

1.5.1 Organizing. I organized 11 conferences: *Math Careers Beyond Academia* (50 participants), *M2@UW* (45 participants), *Geometry & Arithmetic of Surfaces* (40 participants), *CAZoom* (70 participants), *WAGS* (100 participants), *Gems of Combinatorics I & II* (40 & 30 participants), *Spec($\mathbb{Q}$)* (50 participants), *BATMOBILE (2022)* (20 participants), *Gems of Commutative Algebra I & II* (35 & 40 participants). In this time I organized three special sessions at AMS Sectional Meetings and the Joint Math Meetings. Many of these workshops, namely the *Gems of* series, focused on supporting the next generation of researchers by building community and networks.

1.5.2 Math Circles. From 2015 through 2018 I was heavily involved in the Madison Math Circle, including 2 years as the lead organizer. As an organizer, I worked to build stronger connections between the Madison Math Circle, local schools, teachers, and other outreach organizations. This led to weekly attendance to more than double. I also created a new outreach arm of the circle, which visits high schools around the state of Wisconsin to better serve students. While a post-doc at UC Berkeley I volunteered as a speaker with the Berkeley Math Circle.

1.5.3 Mentoring. While a graduate student and post-doc I led four reading courses with undergraduates in algebraic and geometry, volunteered for *Girls' Math Night Out* to promote mathematics to high schoolers, and mentored 6+ students via the AWM's Mentoring Network. During Summer 2021 I led 6 undergraduates on a research project related to [15]. In Summer 2022 I advised an undergraduate student – now a math Ph.D. student with an NSF GRF – on a research project related to the work in Section 1.2.

1.5.4 Virtual Mathematics. In response to the COVID-19 pandemic and the shift of many mathematical activities to virtual formats, I worked to find ways for these online activities to reach those often at the periphery. During Summer and Fall 2020, I helped with Ravi Vakil's *Algebraic Geometry in the Time of Covid* project. This massive online open-access course in algebraic geometry brought together $\sim 2,000$ participants from around the world. In Spring 2021, I organized an 8-week virtual reading course for undergraduates in algebraic geometry and commutative algebra.

1.5.5 Community. While a graduate student I co-founded oSTEM@UW as a resource for students in STEM, and organized a peer networking event *qGrads*, which had over 350 members. During 2020-2023 I served on the board of *Spectra*, including as its inaugural president. I oversaw the growth and formalization of the organization, including the creation and adoption of bylaws, the creation of an invited lecture at the JMM, and a fundraising campaign that has raised over \$20,000.

2. The Geometry of Multigraded Syzygies on Stacks

Let X be a projective variety and L a very ample line bundle. Considering the embedding of X into projective space $X \hookrightarrow \mathbb{P}H^0(X, L) \cong \mathbb{P}^n$ defined by L , let $S_X = S/I_X$ where $S = \mathbb{C}[x_0, \dots, x_n]$ is the standard graded polynomial ring and I_X is the homogeneous defining ideal of X . The minimal graded free resolution of S_X as an S -module is an exact sequence:

$$0 \longleftarrow S_X \xleftarrow{\epsilon} F_0 \xleftarrow{d_1} \cdots \cdots \xleftarrow{d_{k-1}} F_{k-1} \xleftarrow{d_k} F_k \longleftarrow \cdots$$

where F_p is a graded free S -module, and the differentials satisfy some minimality conditions. Since F_p is a graded free S -module it can be written as $\bigoplus_q S(-q)^{\beta_{p,q}(S_X)}$. The module $S(-q)$ is the ring S with a twisted grading, so that $S(-q)_d$ is equal to S_{d-q} where S_{d-q} is the graded piece of degree $d - q$. The $\beta_{p,q}(S_X)$ are known as the graded Betti numbers (or syzygies) of the pair (X, L) , and

we often denote them as $\beta_{p,q}(X; L)$ to indicate their dependence not just on X but on the line bundle L as well. That is to say the syzygies depend on the extrinsic embedding of X into $\mathbb{P}^n$. An overarching theme in the study of syzygies in algebraic geometry is understanding ways that minimal graded free resolutions and graded Betti numbers also capture the intrinsic geometry of X .

An example of this is Mumford's work showing varieties of high degree are defined by quadrics [96]. The connection between the geometry of X and its minimal graded free resolutions has been made most precise when X is a curve. For example, consider the following classical theorem of Noether and Petri concerning canonical curves.

Theorem 2.1 (Noether-Petri Theorem). *Let X be a smooth projective curve of genus g . If X does not have a degree 2 cover of $\mathbb{P}^1$ then the canonical bundle K_X defines a projectively normal embedding of X into $\mathbb{P}^{g-1}$. Further, if X does not admit a degree 3 cover of $\mathbb{P}^1$ then the image of $X \subset \mathbb{P}^{g-1}$ is defined by quadrics.*

In the language of minimal graded free resolutions this theorem can be translated as follows: i) If X does not admit a cover of $\mathbb{P}^1$ of degree 2 then $\beta_{0,0}(X; K_X) = 1$ and $\beta_{0,q}(X; K_X) = 0$ for all $q \neq 0$, and ii) If X does not admit a cover of $\mathbb{P}^1$ of degree 3 then $\beta_{1,q}(X; K_X) = 0$ for all $q \neq 2$. Other results in this vein for curves include Castelnuovo's theorem that if $\deg(L) \geq 2g + 1$ then L is very ample and if $\deg(L) \geq 2g + 2$ then I_X is generated by quadrics, and the theorem that if $\deg(L) \geq 2g + 1$ then S_X is Cohen-Macaulay.

2.1 N_p -Conditions for Stacky Curves A *stacky curve* is a smooth proper geometrically connected Deligne-Mumford stack of dimension 1 which contains a dense open subscheme. Effectively a stacky curve is a curve with a finite number of special points – called stacky points – which have “fractional degrees”. Note this intuition can be made precise in the sense that any stacky curve is (Zariski) locally the quotient of a smooth affine curve by a finite group [103, Lemma 5.3.10]. For simplicity we work over $\mathbb{C}$, however, everything remains true over an arbitrary field if one places minor conditions on the stacky curve.

In [103] Voight and Zureick-Brown develop a theory of line bundles on stacky curves by transferring properties of divisors from their smooth coarse spaces. The key thing to note is that the degree of a line bundle on a stacky curve may be a rational number. Associated to a stacky curve $\mathcal{X}$ and a line bundle $\mathcal{L}$ the *section module* of $\mathcal{X}$ with respect to $\mathcal{L}$ is $R(\mathcal{X}; \mathcal{L}) := \bigoplus_{k \in \mathbb{Z}} H^0(\mathcal{X}, k\mathcal{L})$.

A fundamental difference from the classical theory is that $\mathcal{L}$ provides an embedding into a weighted projective stack, not standard projective space [13]. Algebraically this means that $R(\mathcal{X}; \mathcal{L})$ is a module over a polynomial ring with a non-standard $\mathbb{Z}$ -grading. Assuming $\mathcal{L}$ is sufficiently positive $R(\mathcal{X}; \mathcal{L}) \cong S/I$ for a non-standard $\mathbb{Z}$ -graded polynomial ring $S = \mathbb{C}[x_1, x_2, \dots, x_n]$ where $\deg(x_i) = w_i \in \mathbb{Z}$ and $I \subset S$ some homogeneous ideal.

Example 2.2. Suppose $\mathcal{X}$ is a stacky curve with two stacky points each having $\mathbb{Z}/2\mathbb{Z}$ as its stabilizer and whose coarse space has genus 1. The canonical ring of $\mathcal{X}$ can be minimally presented as

$$R(\mathcal{X}; K_{\mathcal{X}}) \cong \mathbb{C}[x_1, x_2, x_3] / \langle x_3^2 x_1 - b x_3 x_2^6 - x_1^2 - a x_2^8 \rangle$$

where $a, b \in \mathbb{C}$ and $\deg(x_1) = 4$, $\deg(x_2) = 1$, and $\deg(x_3) = 2$ [103, Remark 8.3.6]

Research Goal 2.3. *Understand the syzygies of $R(\mathcal{X}, \mathcal{L})$ where $\mathcal{X}$ is a stacky curve and $\mathcal{L}$ is a line bundle on $\mathcal{X}$.*

The heuristic for this goal is two-fold: i) the syzygies of section rings of stacky curves should behave roughly like the syzygies of weighted log complete series on a curves (see [18]), and ii) Stacky-ness should be thought of as increasing the positivity of line bundle. Justification for the first heuristic comes in the form of the following lemma.

Lemma 2.4. *Let $\mathcal{X}$ be a stacky curve with $\pi : \mathcal{X} \rightarrow X$ its coarse space. If $\mathcal{L}$ is a line bundle on $\mathcal{X}$ then there exists a weighted log incomplete series W on X such that $R(\mathcal{X}; \mathcal{L}) \cong \bigoplus_{d \in \mathbb{Z}} H^0(X, dW)$.*

Justification for the second heuristic is less precise, however, if $\mathcal{L}$ is a line bundle on $\mathcal{X}$ then $\deg(\mathcal{L})$ is equal to $\deg(L) + \sum_i (1 - \frac{1}{e_i})$ where L is a line bundle on X and e_i are the order of the stabilizers of the stacky points. Thus, $\deg(\mathcal{L})$ is “increased” by the presence of stacky points.

Research Problem 2.5. *Let $\mathcal{X}$ be a stacky curve with e the maximum order of the stabilizer groups of stacky points on $\mathcal{X}$. Let $\mathcal{L}$ a line bundle on $\mathcal{X}$.*

- (1) *Find a characterization in terms of $\deg(\mathcal{L})$ for when $R(\mathcal{X}; \mathcal{L})$ is Cohen-Macaulay.*
- (2) *Find a characterization in terms of $\deg(\mathcal{L})$ for when $R(\mathcal{X}; \mathcal{L})$ is generated in degrees $\leq 2e$.*

Brown and Erman recently introduced an analog of Green’s N_p conditions for varieties embedded in weighted projective space [18, Definition 1.2]. Given this, I would like to understand which N_p properties stacky curves satisfy.

Research Problem 2.6. *Let $\mathcal{X}$ be a stacky curve whose coarse space has genus 0 or 1. If $\mathcal{L}$ is a line bundle on $\mathcal{X}$ and $\deg(\mathcal{L}) \gg 0$, what N_p conditions does $R(\mathcal{X}; \mathcal{L})$ satisfy? Determine precise bounds on $\deg(\mathcal{L})$ guaranteeing N_p .*

We conjecture that $R(\mathcal{X}; \mathcal{L})$ satisfies higher N_p conditions that grow roughly linearly with $\deg(\mathcal{L})$. In ongoing work with Maya Banks and Christopher Manon, we are generalizing Green’s kernel bundle techniques [61] to prove this and resolve Problem 2.6. We have successfully constructed a multigraded generalization of these bundles whose cohomology computes the multigraded Betti numbers of $R(\mathcal{X}; \mathcal{L})$. Our current objective is to prove the necessary vanishing theorems for these bundles, which would establish the conjecture; even just under the assumptions the coarse space of $\mathcal{X}$ has genus 0 or 1.

2.2 Green’s Conjecture for Canonical Stacky Curves One perspective on the Noether-Petri Theorem is that the syzygies of a smooth projective curve X seem to be closely related to the degrees for which X admits a cover of $\mathbb{P}^1$. This heuristic turns out to be true in several ways, but to understand the higher syzygies of canonical curves one needs a slightly more subtle invariant. The Clifford index of a smooth curve X is defined to be:

$$\text{Cliff}(X) := \min \left\{ \deg(L) - 2 \dim H^0(X, L) + 2 \mid \begin{array}{l} L \in \text{Pic}(X) \\ \deg(L) \leq g - 1 \quad \dim H^0(X, L) \geq 2 \end{array} \right\}.$$

As a vast generalization of the Noether-Petri theorem, Green’s conjecture states that the syzygies of canonical curves (i.e. the syzygies of a (non-hyperelliptic) curve embedded by the complete linear series of the canonical divisor) are controlled by the Clifford index.

Conjecture 2.7 (Green’s Conjecture [61]). *If X is a smooth projective curve then:*

$$\beta_{p, p+2}(X; K_X) = 0 \quad \text{for all} \quad p < \text{Cliff}(X).$$

Stated differently Green’s conjecture say that one can read the Clifford index of a smooth curve (which is an intrinsic invariant of the curve, not dependent on the embedding) from the vanishing of the syzygies of the canonical embedding of the curve (which are extrinsic to the curve, dependent on the choice of the embedding). Breakthrough work of Voisin proved this conjecture for general curves of every genus [104, 105], and in recent years several new proofs have been found [3, 81]. Substantial work has gone into finding refinements and extensions of Green’s Conjecture [52, 33, 50, 51, 44, 82].

Research Goal 2.8. *Formulate and investigate an appropriate generalization of Green’s conjecture describing the vanishing of the syzygies of the canonical rings of stacky curves.*

The first steps towards Goal 2.8 were taken by Voight and Zureick-Brown who proved a stacky analog of the Noether-Petri theorem.

Theorem 2.9. [103, Theorem 8.4.1] *Let $\mathcal{X}$ be stacky curve whose coarse space has genus ≥ 1 and let e be the maximum order of the stabilizer groups of the stacky points; then the canonical ring $R(\mathcal{X}; K_{\mathcal{X}})$ is generated by elements in degree at most $3e$ with relations in degree at most $6e$.*

This theorem shows that complexity is bounded by the stacky structure, but these bounds are likely not sharp. For example, when $e = 1$ this fails to recover the classical result, instead allowing both quadrics and cubics. Further, it does not provide a geometric explanation of the failure of $R(\mathcal{X}; K_{\mathcal{X}})$ to be generated in the smallest single degree. We propose to refine Theorem 2.9 by studying the relationship between weighted linear series on $\mathcal{X}$ and the syzygies of the canonical ring of $\mathcal{X}$. A far-reaching goal of this would be to introduce a notion of Clifford index for stacky curves.

Research Problem 2.10. *Can Theorem 2.9 be refined by considering the existence of certain stacky or weighted linear series on $\mathcal{X}$? Formulate a stacky analogue of Clifford’s index.*

To define a meaningful Clifford index, we need a robust theory of linear series on stacks; including studying the geometry of the stacky Brill-Noether loci $\mathcal{W}_d^r(\mathcal{X})$, parameterizing line bundles of degree d with at least $r + 1$ independent sections, taking the stacky structure into account. The classical Brill-Noether theorem describes the expected dimension, irreducibility and smoothness of these loci for a general curve [62, 53, 57].

Research Problem 2.11. *For a general stacky curve $\mathcal{X}$ compute the expected dimension of $\mathcal{W}_d^r(\mathcal{X})$ in terms of the genus of the coarse space and the stacky invariants, and determine whether the stacky Brill-Noether locus is of the expected dimension, smooth away from $\mathcal{W}_d^{r+1}(\mathcal{X})$ and irreducible when positive dimensional.*

Recent work refines classical Brill–Noether theory by tracking data analogous to stacky structure (e.g., specified ramification) [100, 88, 89, 71, 35, 34, 87]. The resulting toolkit – degenerations, limit linear series, and tropical/logarithmic method, etc. – appears well suited to possible adaption to the stacky setting, and I am optimistic that these techniques can be adapted to address Problem 2.11. As an achievable first step (suitable for a graduate student project), we will resolve this question completely for stacky curves whose coarse space has genus 0 or 1 with at most 3 or 4 stacky points, where explicit computation of $\mathcal{W}_d^r(\mathcal{X})$ is more tractable.

Another approach one may take towards Goal 2.8 is by studying the algebraic properties of the canonical rings $R(\mathcal{X}; K_{\mathcal{X}})$. As a step in this direction the PI and co-author has shown that just like classical curves [102, 99], the canonical ring of stacky curves is generally Gorenstein.

Lemma 2.12. *If $\mathcal{X}$ is a stacky curve whose coarse space is not hyperelliptic, trigonal, or isomorphic to a plane quintic then the canonical ring $R(\mathcal{X}; K_{\mathcal{X}})$ is Gorenstein.*

The proof of this lemma largely follows the classical argument, after passing to the coarse space, with some slight delicateness needed in handling the non-standard grading. A significantly more subtle algebraic question would be to determine when $R(\mathcal{X}; K_{\mathcal{X}})$ is Koszul, in the sense of Koszulness for modules over a non-standard graded polynomial ring developed recently in [43].

Research Problem 2.13. *Let $\mathcal{X}$ be a stacky curve whose coarse space has genus g . For what t does the canonical ring $R(\mathcal{X}; K_{\mathcal{X}})$ satisfy the t -Koszul property above.*

At this time I have no strong guess for what the correct characterization of when $R(\mathcal{X}; K_{\mathcal{X}})$ is t -Koszul should be. However, in exaplace checking t -Koszul-ness can be done computationally.

The proof of Theorem 2.9 builds upon the ideas of Schreyer [101] to show the existence of a Gröbner basis for the canonical ring of a stacky curve whose elements have certain degrees. Voight and Zureick-Brown do not explicitly construct such Gröbner bases, however, this should be possible for simple stacky curves. A starting point towards Goal 2.8 is to make Voight and Zureick-Brown’s arguments explicit, and then use a computer algebra system like Macaulay2 to compute the minimal graded free resolutions for many examples.

Research Problem 2.14. *Compute the minimal graded free resolution of the canonical ring $R(\mathcal{X}, K_{\mathcal{X}})$ for stacky curves $\mathcal{X}$ where the coarse space of $\mathcal{X}$ has genus ≤ 4 , there are at most 4 stacky points, and the stabilizers of the stacky points have order at most 5.*

Building upon code shared by Voight and Zureick-Brown, I have carried out Problem 2.14 for most of the genus zero examples. In many of these computed cases, and as noted in [103, Appendix], the canonical ring is a (weighted) complete intersection. However, there are examples showing that even for small examples the canonical ring of a stacky curve is quite complicated.

Example 2.15. Let $\mathcal{X}$ be a stacky curve with three stacky points whose stabilizers are $\mathbb{Z}/4\mathbb{Z}$, $\mathbb{Z}/4\mathbb{Z}$, and $\mathbb{Z}/5\mathbb{Z}$ whose coarse space has genus zero. The canonical ring $R(\mathcal{X}; K_{\mathcal{X}})$ can be presented as S/I where $S = \mathbb{C}[x_1, x_2, x_3, x_4, x_5]$ where $\deg(x_1) = 3$, $\deg(x_2) = \deg(x_3) = 4$, and $\deg(x_4) = \deg(x_5) = 5$ and I is an ideal generated by 7 homogeneous polynomials of degrees 8, 8, 9, 9, 10, 13, 13. The minimal graded free resolution of $R(\mathcal{X}; K_{\mathcal{X}})$ is shown below (with the degrees suppressed for brevity):

$$0 \longleftarrow R(\mathcal{X}; K_{\mathcal{X}}) \longleftarrow S \longleftarrow S^{\oplus 7} \longleftarrow S^{\oplus 20} \longleftarrow S^{\oplus 30} \longleftarrow S^{\oplus 22} \longleftarrow S^{\oplus 6} \longleftarrow 0.$$

The approach of using large-scale computations to find conjectures concerning syzygies is very much in the same spirit as my prior work on the syzygies of surfaces [25, 23]. While the precise techniques needed to address Problem 2.14 will likely be different, much of the general framework of organizing, distributing, and publicizing such computations will likely be similar.

2.3 Asymptotic Syzygies of Weighted Projective Stacks The study of asymptotic syzygies investigates the behavior of Betti numbers $\beta_{p,q}(X, L_d)$ as the positivity of the embedding line bundle L_d grows. Ein and Lazarsfeld showed that for a smooth projective variety X , the shape of the Betti table stabilizes as $d \rightarrow \infty$ (see Section 1.1). I propose to generalize this theory to smooth DM stacks. Toric stacks, including weighted projective stacks, provide a natural setting for this investigation.

Research Problem 2.16. *Let $\mathcal{X}$ be a smooth toric DM stack, $\dim \mathcal{X} \geq 2$, and fix an index $1 \leq q \leq \dim \mathcal{X}$. If $(\mathcal{L}_d)_{d \in \mathbb{N}}$ is a sequence of very ample line bundles such that $\mathcal{L}_{d+1} - \mathcal{L}_d$ is constant and ample compute the limit (if it exists):*

$$\lim_{d \rightarrow \infty} \rho_q(\mathcal{X}; \mathcal{L}_d) = \lim_{d \rightarrow \infty} \frac{\#\{p \in \mathbb{N} \mid \beta_{p,p+q}(\mathcal{X}, \mathcal{L}_d) \neq 0\}}{n_d}.$$

We will begin by focusing on specific examples, such as weighted projective planes. Let $\mathcal{P}(a, b, c)$ be the quotient stack $(\mathbb{C}^3 \setminus \{0\})/\mathbb{C}^\times$ where $\mathbb{C}^\times$ acts by $\lambda \cdot (x, y, z) = (\lambda^a x, \lambda^b y, \lambda^c z)$. The coarse space of $\mathcal{P}(a, b, c)$ is the usual weighted projective space $\mathbb{P}(a, b, c)$.

Research Problem 2.17. *Fix a positive integer $a > 1$. For what values of p does the weighted projective stacky plane $\mathcal{P}(1, 1, a)$ embedded by $\mathcal{O}(d)$ satisfy or not satisfy N_p .*

The stack $\mathcal{P}(1, 1, a)$ has a single stacky point with stabilizer $\mathbb{Z}/a\mathbb{Z}$. Its Cox ring is $S = \mathbb{K}[x, y, z]$ with $\deg(x) = \deg(y) = 1$ and $\deg(z) = a$. We anticipate that the required degree d for N_p to hold will grow linearly with a . I hope to approach both Problem 2.16 and Problem 2.17 in a somewhat unified method. I believe it will be possible to adopt the methods of constructing explicit non-zero syzygies in the Koszul cohomology after quotienting by a regular sequence developed in [45] and extended by the PI (see Section 1.1 and [22]). Ideally these can be extended asymptotic results for toric stacks by using a weighted version of Noether normalization argument.

2.4 Computations with Toric Stacks The lack of accessible computational tools is a significant bottleneck in the study of homological invariants of stacks. While a general framework is out of reach, toric stacks are tractable: their combinatorics is well developed, they correspond to *stacky fans* – a fan $\Sigma \subset N_{\mathbb{R}}$ and a lattice map $\beta : N \rightarrow L$ – that makes computations feasible [54, 55, 12]. I am not

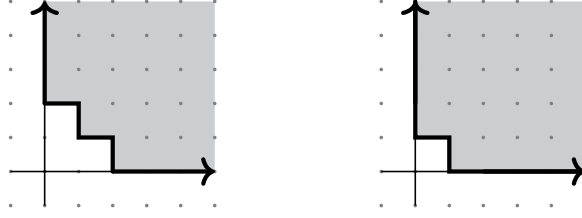


FIGURE 2. Left: The multigraded Castelnuovo–Mumford regularity of $\mathcal{O}_X$ where $X \subset \mathbb{P}^1 \times \mathbb{P}^1$ is the three points $([1 : 1], [1 : 4])$, $([1 : 2], [1 : 5])$, $([1 : 3], [1 : 6])$. Right: The multigraded Castelnuovo–Mumford regularity of $\mathcal{O}_{\mathcal{H}_3}$.

aware of an implementation for working with them. Given the success of the *NormalToricVarieties* package, a *toricStacks* package would benefit both this project and the community.

Research Problem 2.18. *Develop a robust software package for Macaulay2 allowing for computations with toric stacks.*

I will develop the Macaulay2 package *toricStacks*, implementing stacky fans and their morphisms, line bundles on toric stacks, Cox rings, and sheaf cohomology. Work began in Summer 2025 with Maya Banks and John Cobb. I have extensive experience building Macaulay2 packages and will involve graduate students. The package will be open-source.

As computations of examples of syzygies have frequently proven fruitful, I would like to extend the database of syzygies I created in [25, 23].

Research Problem 2.19. *Adapt the high-performance computing methods in [25, 23] to compute the graded Betti numbers of $\mathcal{P}(1, 1, a)$ embedded by $\mathcal{O}(d)$ for $1 < a \leq 15$ and $2 \leq d \leq 6$.*

3. Castelnuovo–Mumford Regularity: Resolutions, Bounds, and Asymptotic Behavior

Building directly on the PI’s work [28, 27] this section outlines the proposal’s second research thrust: characterizing the asymptotic behavior of regularity, finding meaningful bounds on multigraded regularity, and generalizing the Eisenbud-Goto correspondence to the toric setting. See Section 1.2 for a brief introduction to toric varieties, in this section we frequently use the following notation.

Notation 3.1. *We let Y be a smooth projective toric variety coming from a fan $\Sigma \subset N_{\mathbb{R}}$ with $\text{Pic}(Y) \cong \mathbb{Z}^r$. We denote the $\text{Pic}(Y)$ -graded Cox ring of Y by $S = \mathbb{C}[x_1, \dots, x_n]$ and let $B \subset S$ be the irrelevant ideal of Y .*

MacLagan and Smith, defined multigraded regularity for coherent sheaves on toric varieties or equivalently $\mathbb{Z}^r$ -graded modules over the Cox ring S (Definition 1.5). While it serves the classical role of measuring a module’s complexity, multigraded regularity exhibits several new phenomena. For example, $\text{reg}(M)$ is not a number, but a subset of $\mathbb{Z}^r$, and it does not necessarily have a unique minimal element (Figure 3); nor is $\text{reg}(M)$ determined by the multigraded Betti numbers of M [28].

3.1 Asymptotic Behavior and Bounds on Multigraded Regularity In the classical setting of standard graded polynomial rings, significant attention has been given to studying the asymptotic behavior of Castelnuovo–Mumford regularity in various families, e.g., powers of ideals [32, 86, 41, 5] and powers of ideal sheaf [39, 40, 42], and providing meaningful geometric bounds on the regularity of ideals defining smooth subvarieties [64, 48, 10, 94]. The same questions for multigraded regularity remain widely open. The PI hopes to resolve this by building upon her recent work [27]

Research Problem 3.2. Using Notation 3.1, understand the asymptotic behavior of $\text{reg}(I^p)$ and $\text{reg}(\mathcal{I}^p)$ where $I \subset S$ is a $\mathbb{Z}^r$ -graded ideal and $\mathcal{I} \subset \mathcal{O}_Y$ is an ideal sheaf.

Ongoing work of the PI with Lauren Cranton Heller, Mahrud Sayrafi, and Alexandra Seceleanu on led to the following conjecture, which would provide a precise answer to Problem 3.2.

Conjecture 3.3. Let Y be a smooth projective toric variety and $\mathcal{I} \subset \mathcal{O}_Y$ an ideal sheaf. If $\pi : \tilde{Y} \rightarrow Y$ be the blow-up of Y along $\mathcal{I}$ and let E denote the exceptional divisor then:

$$\lim_{p \rightarrow \infty} \frac{\text{reg}(\mathcal{I}^p)_{\mathbb{R}}}{p} = \left\{ L \in \text{Pic}(Y)_{\mathbb{R}} \mid \pi^*(L) - E \text{ is nef on } \tilde{Y} \right\}$$

where the limit is in the sense of Painlevé–Kuratowski convergence of subsets of $\mathbb{R}^n$.

We call the set on the right hand side of Conjecture 3.3 the Seshadri region of $\mathcal{I}$ as it is a natural multigraded analog of Seshadri constants, which measure positivity. One can think of this conjecture as the natural analog of [42]. This conjecture suggests that the asymptotically, multigraded regularity – which is typically complicated and non-convex – is governed by a surprisingly simple, convex geometric object measuring positivity. In ongoing work, the PI has established key steps toward the proof and anticipates resolving this conjecture during the grant period.

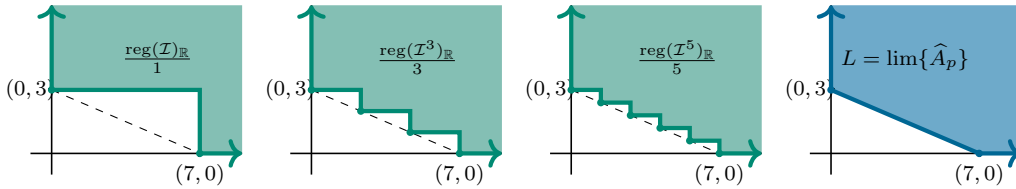


FIGURE 3. The picture of the convergence of $\frac{\text{reg}(\mathcal{I}^p)_{\mathbb{R}}}{p}$ to the Seshadri region of $\mathcal{I}$ predicted Conjecture 3.3.

Given its definition and that it is a measure of positivity for an ideal sheaf, it is natural to wonder whether the Seshadri region of $\mathcal{I}$ predicted Conjecture 3.3 is connected to the theory of Newton-Okounkov bodies, which are also convex bodies roughly measuring positivity of a linear series along a flag of subvarieties that have received significant study [97, 98, 91, 74, 2, 73]

Research Problem 3.4. Are the Seshadri regions Newton-Okounkov bodies for some flag on Y .

Beyond the asymptotics of $\text{reg}(I^p)$ and $\text{reg}(\mathcal{I}^p)$ it would be useful to have a finer understanding of how these regions change with p . Computation and Conjecture 3.3 suggest that the number of minimal generators with respect to the Nef(Y) ordering of $\text{reg}(I^p)$ and $\text{reg}(\mathcal{I}^p)$ increases with p . This growth is a genuinely multigraded phenomenon, as on $\mathbb{P}^n$ regularity is generated by a single element. Quantifying and explaining this growth is an interesting question within reach.

Research Problem 3.5. Using Notation 3.1 let $I \subset S$ be a $\mathbb{Z}^r$ -graded ideal and $\mathcal{I} \subset \mathcal{O}_Y$ an ideal sheaf on Y : (1) Bound the number of minimal generators of $\text{reg}(I^p)$ as a function of p . (2) Bound the number of minimal generators of $\text{reg}(\mathcal{I}^p)$ as a function of p .

My hope for part (2) of Problem 3.5 is to refine the convergence in Conjecture 3.3 to produce a lower bound on the number of minimal generators of $\text{reg}(\mathcal{I}^p)$ in terms of the number of lattice points “near” the Seshadri region of $\mathcal{I}$ via Ehrhart theory.

In a different direction, I will prove meaningful bounds on the regularity of the defining ideals of subvarieties of Y in the vein of classical results like Gruson, Lazarsfeld, Peskine [64], Bertram, Ein, Lazarsfeld [10], and the Eisenbud–Goto conjecture [48] (recently shown false [94]).

Research Problem 3.6. *Using Notation 3.1 let $Z \subset Y$ be smooth subvariety. Bound the multigraded regularity of I_Z and S/I_Z in terms of $\text{codim}(Z \subset Y)$ and the degrees of minimal generators of I_Z ?*

I will approach Problem 3.6 by beginning with understanding the regularity of I_Z when Z is a torus invariant subvariety of Y . In this setting there is a bijection between such subvarieties and cones in the fan defining X , and the defining equations of Z are entirely determined by the combinatorics of the corresponding cone.

Research Problem 3.7. *Compute the multigraded regularity of torus invariant subvarieties of Y in terms of the combinatorics of the cone defining them.*

Despite being a relatively simple class of subvarieties it has been shown that their multigraded regularity behaves in nuanced ways [27]. The goal for approaching Problem 3.7 is to compute both the higher cohomology of $\mathcal{I}_Z$ and its Hilbert function using the combinatorics of the corresponding cone and working inductively on codimension. A degeneration argument and upper semicontinuity could then be used to extend an answer for Problem 3.7 to an answer for Problem 3.6.

Lastly I will pursue the following focused project concerning the defining equations and regularity of the Deligne-Mumford moduli space of n -pointed stable curves of genus zero, denoted by $\overline{M}_{0,n}$, which has rich geometry and combinatorics [85, 56, 79, 72, 84, 93, 30, 59]. Keel and Tevelev constructed an embedding $\Omega_n : \overline{M}_{0,n} \hookrightarrow \mathbb{P}^1 \times \mathbb{P}^2 \times \cdots \times \mathbb{P}^{n-3}$, through which the log canonical embedding given by $\partial\overline{M}_{0,n} + K_{\overline{M}_{0,n}}$ factors [80]. Monin and Rana [95] produced explicit equations, discussed below, which they conjectured defined the scheme-theoretic image of Ω_n . Gillespie, Griffin, and Levinson recently proved this [58].

Research Problem 3.8. *Compute the multigraded Castelnuovo–Mumford Regularity and minimal graded free resolution of the defining ideal of $\Omega_n(\overline{M}_{0,n}) \subset \mathbb{P}^1 \times \mathbb{P}^2 \times \cdots \times \mathbb{P}^{n-3}$.*

Using *Macaulay2*, the regularity and free resolution of $\Omega_n(\overline{M}_{0,n})$ is computable for $n = 4, 5, 6, 7$. The Monin-Rana equations defining $\Omega_n(\overline{M}_{0,n})$ arise as follows: Let $x_0^{(i)}, x_1^{(i)}, \dots, x_i^{(i)}$ denote the coordinates on $\mathbb{P}^i$, then for each $i < j$ let

$$M_{i,j} := \begin{pmatrix} x_0^{(i)}(x_0^{(j)} - x_i^{(j)}) & x_1^{(i)}(x_1^{(j)} - x_i^{(j)}) & x_2^{(i)}(x_2^{(j)} - x_i^{(j)}) & \cdots & x_i^{(i)}(x_{i+1}^{(j)} - x_i^{(j)}) \\ x_0^{(j)} & x_1^{(j)} & x_2^{(j)} & \cdots & x_{i+1}^{(j)} \end{pmatrix}.$$

Define $J_{i,j}$ to be the ideal generated by the 2×2 minors of $M_{i,j}$. The ideal defining $\Omega_n(\overline{M}_{0,n})$ is then $I_n = \sum_{1 \leq i < j \leq n-3} J_{i,j}$. Each ideal $J_{i,j}$ is independent of n and is resolved by an Eagon–Northcott complex. Further the defining ideals of I_n have natural inductive structure since $I_n = I_{n-1} + J_{1,n-3} + \cdots + J_{n-4,n-3}$. Working inductively and using a mapping cone construction over the short exact sequence arising from ideal sums we can produce a non-minimal free resolution of $\Omega_n(\overline{M}_{0,n})$. The key technical step is to control the multigraded resolutions of intersections $J_{i,j} \cap J_{i',j'}$. If we can control these, we can then hope to minimize the resulting resolution to get the minimal free resolution of $\Omega_n(\overline{M}_{0,n})$. This picture is also compatible with the following alternative approach: $\Omega_n(\overline{M}_{0,n})$ is the image of a product of GL -equivariant bundles and so can use Kempf collapsing and the Kempf–Lascoux–Weyman geometric technique for computing syzygies [83, 90, 106], which has recently been applied successfully in similar situations [8].

3.2 A Generalized Eisenbud–Goto Theorem for Toric Varieties The relationship between Castelnuovo–Mumford regularity, graded Betti numbers, and linear resolutions is fundamental in classical commutative algebra (see Theorem 1.4 [48]). In the multigraded the analog of a linear resolution is a resolution such that each twist at homological degree i is in $L_i(\mathbf{0})$, i.e., the entries of the differential have total degree one. However, with these definitions the relationship between linear resolutions and regularity breaks down dramatically.

Example 3.9. Consider $\mathbb{P}^1 \times \mathbb{P}^2$ so that $S = \mathbb{K}[x_0, x_1, y_0, y_1, y_2]$ and $B = \langle x_0, x_1 \rangle \cap \langle y_0, y_1, y_2 \rangle$. One checks that S/B is $(0, 0)$ -regular. The minimal graded free resolution of $(S/B)_{\geq \mathbf{0}} = S/B$ is:

$$S \longleftarrow S(-1, -1)^6 \longleftarrow \begin{matrix} S(-1, -2)^6 \\ \oplus \\ S(-2, -1)^3 \end{matrix} \longleftarrow \begin{matrix} S(-1, -3)^2 \\ \oplus \\ S(-2, -2)^3 \end{matrix} \longleftarrow S(-2, -3) \longleftarrow 0.$$

In particular, S/B does not have a linear resolution as $(-1, -1) \notin L_1(\mathbf{0})$, and so being $\mathbf{d}$ -regular does not imply $M_{\geq \mathbf{d}}$ has linear resolution. In alignment with Theorem 1.7 this resolution is $\mathbf{0}$ -quasilinear.

Examples, like the one above, led the PI and co-authors, to introduce the weaker notion of quasilinear resolutions (see Definition 1.6) which they used to characterize multigraded regularity on products of projective space [28]. The PI plans to generalize these results to other toric varieties.

Research Problem 3.10. *Using Notation 3.1 let M be a $\text{Pic}(Y)$ -graded S -module. Find the correct generalization of d -quasi-linear resolution such that a $\text{Pic}(Y)$ -graded S -module M is d -regular if and only if the truncation $M_{\geq d}$ has a d -quasilinear resolution.*

We will begin with the case of Hirzebruch surfaces $\mathcal{H}_t = \mathbb{P}(\mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(t))$ where the PI has the following precise conjecture.

Conjecture 3.11. *Let S be the $\mathbb{Z}^2$ -graded Cox ring of the Hirzebruch surface $\mathcal{H}_t$. If M is a $\mathbb{Z}^2$ -graded S -module then $M_{\geq d}$ has a d -linear resolution if and only if $d + \text{reg}(S) \subset \text{reg}(M)$.*

The PI's work in [28] suggests a general approach to Problem 3.10 and Conjecture 3.11: i) Find an appropriate collection of line bundles $L_1, \dots, L_t$ on Y in which one can resolve the diagonal in $\Delta \subset Y \times Y$, ii) Understand the image of these line bundles under the Fourier-Mukai transform whose kernel is this resolution of Δ , iii) Prove strong vanishing results for the higher cohomology of the given line bundles L_i . This allows us to construct an explicit resolution of $M_{\geq \mathbf{d}}$ whose graded Betti numbers are given, roughly, by the cohomology of $\tilde{M} \otimes L_i$. Recent work by Hanlon, Hicks, and Lazarev suggest the Bondal-Thomsen collection is a natural choice for the L_i ; as they give a nice explicit resolution of the diagonal of an arbitrary toric variety and generally have strong vanishing properties [67, 17]. Further, some work has gone into step (ii) for this collection [38]. On Hirzebruch surfaces various collections have been studied even further [19].

While not equivalent to being regular, characterizing when $M_{\geq \mathbf{d}}$ has a linear resolution in the traditional sense is still of interest algebraically.

Research Problem 3.12. *Using Notation 3.1 let M be a $\text{Pic}(Y)$ -graded S -module. Characterize when $M_{\geq \mathbf{d}}$ has a linear resolution in terms of vanishing of the local cohomology groups $H_B^i(M)_t$.*

The PI answered this question for products of projective spaces [28, Theorem 6.2]. Using the same methods of controlling the Betti numbers needed to answer Problem 3.10 would likely provide a way to give a cohomological characterization for when $M_{\geq \mathbf{d}}$ has a linear resolution.

Direct computation of the regularity of a module M is often prohibitively slow due to the complexity of local cohomology. A significantly faster approach involves computing the truncation $M_{\geq \mathbf{d}}$ and its associated minimal free resolution. Consequently, Theorem 1.7, combined with the resolution of Conjecture 3.11 and Problem 3.10, yields a practical algorithm for determining regularity at a single multidegree $\mathbf{d} \in \mathbb{Z}^r$. However, developing an algorithm to compute the entire regularity region of M remains a major open challenge.

Research Problem 3.13. *Find an algorithm to compute a minimal generating set for $\text{reg}(M)$.*

This question is open in all cases except $Y = \mathbb{P}^n$. The primary obstacle is determining a box guaranteed to contain all minimal generators of $\text{reg}(M)$. Current bounds typically resemble L-shaped regions (i.e., translates of $\text{Nef}(Y)$), which do not sufficiently constrain the search space for these

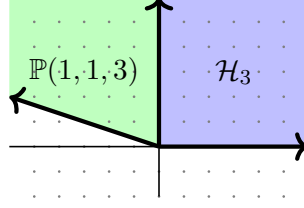


FIGURE 4. The secondary fan of the Hirzebruch surface $\mathcal{H}_3$, which has two chambers; (blue) corresponding to $\mathcal{H}_3$ and (green) corresponding to $\mathbb{P}(1, 1, 3)$.

generators. Therefore, addressing Problem 3.13 is intrinsically linked to this project's objective of establishing tighter bounds on multigraded regularity.

Finally, as noted above, my previous work shows that the multigraded Betti numbers of a module do not determine its multigraded regularity. One might hope this can be saved by considering the multigraded Betti numbers of all virtual resolutions of M . A virtual resolution of a module M is a free $\mathbb{Z}^r$ -graded complex of S -modules $F_\bullet$ such that the corresponding complex $\tilde{F}_\bullet$ of locally free $\mathcal{O}_Y$ -modules is a resolution of the sheaf $\tilde{M}$ [9, Definition 1.1].

Research Problem 3.14. *Using Notation 3.1 let M be a $\text{Pic}(Y)$ -graded S -module. Does the graded Betti numbers of all virtual resolutions of M determine the multigraded regularity of M ?*

4. Uniform Homological Algebra on Toric Varieties

A distinctive feature of toric geometry, stemming from the GIT presentation $Y = (\mathbb{C}^n \setminus \mathbb{V}(B))/(\mathbb{C}^\times)^r$, is that many non-isomorphic toric varieties share the same graded Cox ring $S = \mathbb{C}[x_1, \dots, x_n]$. Fixing a $\mathbb{Z}^r$ -graded ring S determines the torus action of $(\mathbb{C}^\times)^r$ on $\mathbb{C}^n$, but it does not determine a unique geometric realization; different choices of stability conditions, or equivalently irrelevant ideals, give different projective toric varieties with Cox ring S .

The space of all such GIT models is controlled by the secondary (GKZ) fan Σ_{gkz} . A choice of a maximal cone (chamber) Γ in Σ_{gkz} is equivalent to specifying: i) a toric variety Y_Γ with Cox ring S , ii) an irrelevant ideal $B_\Gamma \subset S$, or iii) a GIT stability condition determining the semi-stable locus of the quotient. Combinatorially, one may think of Σ_{gkz} as parameterizing the regular sub-division of a given set of rays. Moreover, Σ_{gkz} governs the birational geometry of these models: crossing a wall between adjacent chambers changes stability and the corresponding varieties are related by an explicit birational transformation (flips, flops), as established in VGIT [37]

Example 4.1. Consider the case of the Hirzebruch surface $\mathcal{H}_3$. Here the Cox ring is $S = \mathbb{C}[x_1, x_2, x_3, x_4]$ with the $\mathbb{Z}^2$ -grading given by $\deg(x_1) = \deg(x_3) = (1, 0)$, $\deg(x_4) = (0, 1)$, and $\deg(x_2) = (-3, 1)$. The secondary/GKZ fan of this Cox ring is shown in Figure 4, and has two chambers one corresponding to $\mathcal{H}_3$ and one corresponding to the weighted projective plane $\mathbb{P}(1, 1, 3)$. The irrelevant ideals are $B_{\mathcal{H}_3} = \langle x_1, x_3 \rangle \cap \langle x_2, x_4 \rangle$ and $B_{\mathbb{P}(1,1,3)} = \langle x_1, x_3, x_4 \rangle \cap \langle x_2 \rangle$.

A developing theme is that many homological properties of toric varieties depend primarily on the graded ring S rather than on a specific model Y [76, 77, 66, 6, 67, 7]. Thus, where the classical algebra–geometry dictionary on $\mathbb{P}^n$ fails to generalize, it suggests the need for a dictionary that simultaneously accounts for all VGIT quotients of a fixed ring S . These issues are invisible on $|\mathbb{P}^n$ because the secondary fan of $\mathbb{P}^n$ has a single chamber. This project aims to establish a uniform framework for studying homological invariants across different realizations of the same Cox ring.

4.1 Regularity on the D_{Cox} Category A recent instantiation of these ideas is the work of [7] showing that birational variant of King’s conjecture on the existence of full strong exceptional collections of line bundles, is true, if one is willing to work on unifying derived category called $D_{\text{Cox}}(Y)$, called the Cox category of Y . The Cox category of Y is the smallest derived category containing the derived category $D(\mathcal{Y}_i)$ for each toric stack $\mathcal{Y}_i$ corresponding to chambers in the secondary fan of Y . While $D_{\text{Cox}}(Y)$ is *not* the derived category of any stack, it has many nice properties. In particular, the Bondal–Thomsen collection is a full strong exceptional collection on $D_{\text{Cox}}(Y)$ and there are natural localization maps from $D_{\text{Cox}}(Y)$ to each VGIT quotient [7].

Research Problem 4.2. *Develop a theory of Castelnuovo–Mumford regularity on the category D_{Cox} that restricts to multigraded regularity on each Y_i .*

One natural approach to developing such a theory is to note that $D_{\text{Cox}}(Y)$ can be realized as an explicit full subcategory of $D(\tilde{\mathcal{Y}})$ where $\tilde{\mathcal{Y}}$ is any smooth toric stack with proper birational morphisms to the $\mathcal{Y}_i$. One could thus, hope to using the notion of multigraded regularity on $\tilde{\mathcal{Y}}$ and restrict it to the $D_{\text{Cox}}(Y)$ subcategory. A main obstruction is that the choice of $\tilde{\mathcal{Y}}$ is not unique, and thus, defining regularity in this way may depend on the choice of $\tilde{\mathcal{Y}}$. I hope to circumventing this by showing that there is, in a sense, a best choice for $\tilde{\mathcal{Y}}$; namely the toric stack $\mathcal{Y}_{\text{gkz}}$ constructed in [7, Section 3] by gluing the stack corresponding to each chamber of Σ_{gkz} . While not unique to $D_{\text{Cox}}(Y)$ this construction of a $\tilde{\mathcal{Y}}$ should be universal in some sense that seems missing from [7, Section 3].

Research Problem 4.3. *Characterize the universality of the $\mathcal{Y}_{\text{gkz}}$ stack construction.*

Proving a strong version of universality would provide a strong step towards defining regularity on $D_{\text{Cox}}(Y)$ via $D(\mathcal{Y}_{\text{gkz}})$. As an effort in this direction, and with aim of making explicit some of the steps of the construction provided in [7], I plan to implement this construction in *Macaulay2*.

Research Problem 4.4. *Implement the construction of the $\mathcal{X}_{\text{gkz}}$ stack in the *toricStacks* package being developed for *Macaulay2* as part of Problem 2.18.*

Work on this problem is ongoing with Maya Banks and John Cobb. We have rough code that can compute $\mathcal{X}_{\text{gkz}}$ in a narrow number of cases, and we are working to expand its functionality.

4.2 Homological Invariants and Wall Crossings An alternative approach of studying homological algebra on all toric varieties with the same Cox ring simultaneously is to understand how things like minimal free resolutions and multigraded regularity change when moving between adjacent chambers Γ and Γ' of the secondary fan. Geometrically, this corresponds to a birational transformation between the toric varieties Y_Γ and $Y_{\Gamma'}$, (e.g. a flip or flop). Algebraically this corresponds to the irrelevant ideal changing B_Γ to $B_{\Gamma'}$.

Geometrically we can think of sheaves crossing between Γ and Γ' by the pushing or pulling along the corresponding birational transformation $Y_\Gamma \rightarrow Y_{\Gamma'}$. Algebraically, this roughly corresponds to fixing a $\mathbb{Z}^r$ -graded module M on the ring S and then saturating with respect to B_Γ or $B_{\Gamma'}$ to get modules M_Γ and $M_{\Gamma'}$ corresponding to sheaves on Y_Γ and $Y_{\Gamma'}$ respectively. A *wall-crossing formula* seeks to relate the algebraic or geometric invariants of M_Γ and $M_{\Gamma'}$. We aim to establish precise wall-crossing formulas for multigraded regularity and multigraded Betti numbers.

Research Problem 4.5. *Fix $I \subset S$, a $\mathbb{Z}^r$ -graded ideal and Γ and Γ' be adjacent chambers in Σ_{gkz} .*

- (1) *(Wall-Crossing for Regularity): Prove there exists a “small” subset $F \subset \mathbb{Z}^r$ independent of I such that:*

$$\text{reg}_{Y_\Gamma}((I :_S B_\Gamma^\infty)) + F = \text{reg}_{Y_{\Gamma'}}((I :_S B_{\Gamma'}^\infty)) + F.$$

- (2) *(Wall-Crossing for Betti Numbers): Prove the minimal graded free resolutions of $(I :_S B_\Gamma^\infty)$ and $(I :_S B_{\Gamma'}^\infty)$ agree in small degree.*

Proving these would establish that the homological complexity does not change arbitrarily under birational transformations, but rather follows predictable patterns dictated by VGIT. The strategy is to exploit the combinatorial control one over the irrelevant ideal during a wall-crossing, which corresponds to a bistellar flip that exchanges two minimal primes in the ideals B_Γ and $B_{\Gamma'}$ [37, Theorem 15.3.13]. A Mayer–Vietoris type spectral sequence can then relate the local cohomology modules $H_{B_\Gamma}^i((I :_S B_\Gamma^\infty))$ and $H_{B_{\Gamma'}}^i((I :_S B_{\Gamma'}^\infty))$ providing some control over the regularity and minimal free resolutions. This approach is further motivated by the fact these birational maps often induce equivalences on derived categories [75, 76, 77, 66, 78, 6]. As a test case for Problems 4.5, I explore varieties Y with Picard rank two, where the secondary fan is two-dimensional.

5. Broader Impacts

5.1 Organizing In addition to the 10+ conferences noted in Section 1.5.1, since starting at Dartmouth in 2024 I organized two algebraic geometry conferences: *AGNES* (150 participants) and *BATMOBILE* (50 participants). I am organizing this year’s *BATMOBILE*, to occur at Amherst College in November 2025, highlighting recent work in combinatorial algebraic geometry and commutative algebra. I am preparing a BIRS proposal titled *The Geometry of Syzygies: Past & Future* and plan to organize the next *Gems of Commutative Algebra* at Dartmouth in Winter 2027 to promote community among the next generation of researchers; if successful, I will write a short Notices of the AMS article on these efforts and how to organize similar conferences amid current logistical hurdles. I am an organizer of Dartmouth’s *Algebra & Number Theory* seminar.

5.2 Campus Community. In an effort to promote community among math majors on Dartmouth’s campus, I serve as the faculty advisor to the college’s *AWM Student Chapter* and am also working with the campus’s *oSTEM* chapter to broaden participation in STEM. Beginning in Fall 2025 I am organizing a graduate student seminar *Unwritten Rules* providing professional development to graduate students in the department facing additional career challenges.

5.3 National & International Advocacy. I have worked with the *AWM* Executive Committee to expand support for broader groups of mathematicians and plan to continue this effort. From Winter 2023 to Spring 2025 I served on SLMath’s *Committee on Women in Mathematics* until its dissolution. I have since agreed to serve on two *AMS* committees to promote mathematics and the community. I hope to resume work with *Spectra*, where I previously served as inaugural president.

5.4 Mentoring. I am mentoring two graduate students, Bailee Z. (Michigan) and Jacob B. (KSU), on a tropical geometry project begun at MREG 2023; a preprint is expected soon. Since Fall 2024, I’ve mentored two undergraduates: Joanna S. (algebra reading course) and Lisa S. (exploring math graduate school). Together with Asher Auel I am co-advising one graduate student, Will D., working on using free resolutions of points on surfaces to study the geometry of the moduli of spin curves. I am pre-advising three students (Allison T., Alex M., Aidan H.) interested in combinatorics/algebra. With these students I am developing a reading course in algebraic geometry aimed at increasing retention among students who are often pushed away from the field, by taking a humanistic approach, combining active learning, exercises in algebraic geometry, and approaches to community building. I will make the notes I am writing for this available on my website. I expect to become the advisor for at least one of these students. I’ve also agreed to lead an undergraduate summer research project at *Count Me In* (organized in part by Milena Hering and Diane MacLagan).

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SUMMARY PROPOSAL BUDGET

YEAR 1

ORGANIZATION Dartmouth College				FOR NSF USE ONLY		
				PROPOSAL NO.	DURATION (months)	
PRINCIPAL INVESTIGATOR / PROJECT DIRECTOR Juliette Bruce				AWARD NO.		
A. SENIOR/KEY PERSONNEL: PI/PI, Co-PI's, Faculty and Other Senior/ Key Associates (List each separately with title, A.7. show number in brackets)				NSF Funded Person-months		Funds Requested By proposer
				CAL	ACAD	
1. Juliette Bruce - Principal Inv				2.0		26,507
2.						
3.						
4.						
5.						
6. () OTHERS (LIST INDIVIDUALLY ON BUDGET JUSTIFICATION PAGE)				0.0		0
7. (1) TOTAL SENIOR/KEY PERSONNEL (1 - 6)				2.0		26,507
B. OTHER PERSONNEL (SHOW NUMBERS IN BRACKETS)						
1. (0) POST DOCTORAL SCHOLARS				0.0		0
2. (0) OTHER PROFESSIONALS (TECHNICIAN, PROGRAMMER, ETC.)				0.0		0
3. (1) GRADUATE STUDENTS						50,427
4. (1) UNDERGRADUATE STUDENTS						1,693
5. (0) SECRETARIAL - CLERICAL (IF CHARGED DIRECTLY)						0
6. (0) OTHER						0
TOTAL SALARIES AND WAGES (A + B)						78,627
C. FRINGE BENEFITS (IF CHARGED AS DIRECT COSTS)						9,297
TOTAL SALARIES, WAGES AND FRINGE BENEFITS (A + B + C)						87,924
D. EQUIPMENT (LIST ITEM AND DOLLAR AMOUNT FOR EACH ITEM EXCEEDING \$5,000.)						
TOTAL EQUIPMENT						0
E. TRAVEL 1. DOMESTIC (INCL. U.S. POSSESSIONS)						6,176
2. INTERNATIONAL						5,000
F. PARTICIPANT SUPPORT COSTS						
1. STIPENDS \$ 0						
2. TRAVEL 0						
3. SUBSISTENCE 0						
4. OTHER 0						
TOTAL NUMBER OF PARTICIPANTS (0) TOTAL PARTICIPANT COSTS						0
G. OTHER DIRECT COSTS						
1. MATERIALS AND SUPPLIES						0
2. PUBLICATION COSTS/DOCUMENTATION/DISEMINATION						0
3. CONSULTANT SERVICES						0
4. COMPUTER SERVICES						5,000
5. SUBAWARDS						0
6. OTHER						5,282
TOTAL OTHER DIRECT COSTS						10,282
H. TOTAL DIRECT COSTS (A THROUGH G)						109,382
I. INDIRECT COSTS (F&A)(SPECIFY RATE AND BASE) MTDC (Rate: 63.5, Base:109382)						
TOTAL INDIRECT COSTS (F&A)						69,458
J. TOTAL DIRECT AND INDIRECT COSTS (H + I)						178,840
K. FEE						0
L. AMOUNT OF THIS REQUEST (J) OR (J MINUS K)						178,840
M. COST SHARING PROPOSED LEVEL \$ 0				AGREED LEVEL IF DIFFERENT \$		
PI/PI NAME Juliette Bruce				FOR NSF USE ONLY		
ORG. REP. NAME*				INDIRECT COST RATE VERIFICATION		
				Date Checked	Date Of Rate Sheet	Initials - ORG

*ELECTRONIC SIGNATURES REQUIRED FOR REVISED BUDGET

SUMMARY PROPOSAL BUDGET

YEAR 2

ORGANIZATION Dartmouth College				FOR NSF USE ONLY		
PRINCIPAL INVESTIGATOR / PROJECT DIRECTOR Juliette Bruce				PROPOSAL NO.	DURATION (months)	
				AWARD NO.	Proposed	Granted
A. SENIOR/KEY PERSONNEL: PI/PD, Co-PI's, Faculty and Other Senior/ Key Associates (List each separately with title, A.7. show number in brackets)				NSF Funded Person-months		Funds Requested By proposer
				CAL	ACAD	SUMR
1. Juliette Bruce - Principal Inv				2.0		27,832
2.						
3.						
4.						
5.						
6. () OTHERS (LIST INDIVIDUALLY ON BUDGET JUSTIFICATION PAGE)				0.0		0
7. (1) TOTAL SENIOR/KEY PERSONNEL (1 - 6)				2.0		27,832
B. OTHER PERSONNEL (SHOW NUMBERS IN BRACKETS)						
1. (0) POST DOCTORAL SCHOLARS				0.0		0
2. (0) OTHER PROFESSIONALS (TECHNICIAN, PROGRAMMER, ETC.)				0.0		0
3. (1) GRADUATE STUDENTS						52,948
4. (1) UNDERGRADUATE STUDENTS						1,778
5. (0) SECRETARIAL - CLERICAL (IF CHARGED DIRECTLY)						0
6. (0) OTHER						0
TOTAL SALARIES AND WAGES (A + B)						82,558
C. FRINGE BENEFITS (IF CHARGED AS DIRECT COSTS)						9,762
TOTAL SALARIES, WAGES AND FRINGE BENEFITS (A + B + C)						92,320
D. EQUIPMENT (LIST ITEM AND DOLLAR AMOUNT FOR EACH ITEM EXCEEDING \$5,000.)						
TOTAL EQUIPMENT						0
E. TRAVEL 1. DOMESTIC (INCL. U.S. POSSESSIONS)						6,176
2. INTERNATIONAL						5,000
F. PARTICIPANT SUPPORT COSTS						
1. STIPENDS \$ <u>0</u>						0
2. TRAVEL <u>0</u>						0
3. SUBSISTENCE <u>0</u>						0
4. OTHER <u>0</u>						0
TOTAL NUMBER OF PARTICIPANTS (0) TOTAL PARTICIPANT COSTS						0
G. OTHER DIRECT COSTS						
1. MATERIALS AND SUPPLIES						0
2. PUBLICATION COSTS/DOCUMENTATION/DISSEMINATION						0
3. CONSULTANT SERVICES						0
4. COMPUTER SERVICES						0
5. SUBAWARDS						0
6. OTHER						5,546
TOTAL OTHER DIRECT COSTS						5,546
H. TOTAL DIRECT COSTS (A THROUGH G)						109,042
I. INDIRECT COSTS (F&A)(SPECIFY RATE AND BASE) MTDC (Rate: 63.92, Base:109042)						
TOTAL INDIRECT COSTS (F&A)						69,700
J. TOTAL DIRECT AND INDIRECT COSTS (H + I)						178,742
K. FEE						0
L. AMOUNT OF THIS REQUEST (J) OR (J MINUS K)						178,742
M. COST SHARING PROPOSED LEVEL \$ 0				AGREED LEVEL IF DIFFERENT \$		
PI/PD NAME Juliette Bruce				FOR NSF USE ONLY		
ORG. REP. NAME*				INDIRECT COST RATE VERIFICATION		
				Date Checked	Date Of Rate Sheet	Initials - ORG

*ELECTRONIC SIGNATURES REQUIRED FOR REVISED BUDGET

SUMMARY PROPOSAL BUDGET

Cumulative

ORGANIZATION Dartmouth College				FOR NSF USE ONLY		
PRINCIPAL INVESTIGATOR / PROJECT DIRECTOR Juliette Bruce				PROPOSAL NO.	DURATION (months)	
				AWARD NO.	Proposed	Granted
A. SENIOR/KEY PERSONNEL: PI/PD, Co-PI's, Faculty and Other Senior/Key Associates (List each separately with title, A.7. show number in brackets)				NSF Funded Person-months		Funds Requested By proposer
				CAL	ACAD	SUMR
1. Juliette Bruce - Principal Inv				4.0		54,339
2.						
3.						
4.						
5.						
6. () OTHERS (LIST INDIVIDUALLY ON BUDGET JUSTIFICATION PAGE)						
7. (1) TOTAL SENIOR/KEY PERSONNEL (1 - 6)				4.0		54,339
B. OTHER PERSONNEL (SHOW NUMBERS IN BRACKETS)						
1. (0) POST DOCTORAL SCHOLARS				0.0		0
2. (0) OTHER PROFESSIONALS (TECHNICIAN, PROGRAMMER, ETC.)				0.0		0
3. (2) GRADUATE STUDENTS						103,375
4. (2) UNDERGRADUATE STUDENTS						3,471
5. (0) SECRETARIAL - CLERICAL (IF CHARGED DIRECTLY)						0
6. (0) OTHER						0
TOTAL SALARIES AND WAGES (A + B)						161,185
C. FRINGE BENEFITS (IF CHARGED AS DIRECT COSTS)						19,059
TOTAL SALARIES, WAGES AND FRINGE BENEFITS (A + B + C)						180,244
D. EQUIPMENT (LIST ITEM AND DOLLAR AMOUNT FOR EACH ITEM EXCEEDING \$5,000.)						
TOTAL EQUIPMENT						0
E. TRAVEL 1. DOMESTIC (INCL. U.S. POSSESSIONS)						12,352
2. INTERNATIONAL						10,000
F. PARTICIPANT SUPPORT COSTS						
1. STIPENDS \$ <u>0</u>						0
2. TRAVEL <u>0</u>						0
3. SUBSISTENCE <u>0</u>						0
4. OTHER <u>0</u>						0
TOTAL NUMBER OF PARTICIPANTS (0) TOTAL PARTICIPANT COSTS						0
G. OTHER DIRECT COSTS						
1. MATERIALS AND SUPPLIES						0
2. PUBLICATION COSTS/DOCUMENTATION/DISSEMINATION						0
3. CONSULTANT SERVICES						0
4. COMPUTER SERVICES						5,000
5. SUBAWARDS						0
6. OTHER						10,828
TOTAL OTHER DIRECT COSTS						15,828
H. TOTAL DIRECT COSTS (A THROUGH G)						218,424
I. INDIRECT COSTS (F&A)(SPECIFY RATE AND BASE)						
TOTAL INDIRECT COSTS (F&A)						139,158
J. TOTAL DIRECT AND INDIRECT COSTS (H + I)						357,582
K. FEE						0
L. AMOUNT OF THIS REQUEST (J) OR (J MINUS K)						357,582
M. COST SHARING PROPOSED LEVEL \$ 0				AGREED LEVEL IF DIFFERENT \$		
PI/PD NAME Juliette Bruce				FOR NSF USE ONLY		
ORG. REP. NAME*				INDIRECT COST RATE VERIFICATION		
				Date Checked	Date Of Rate Sheet	Initials - ORG

*ELECTRONIC SIGNATURES REQUIRED FOR REVISED BUDGET

BUDGET JUSTIFICATION

Dartmouth College defines a year as 12 calendar months. Salaries for all personnel are increased by 5% annually effective July 1st at the beginning of Dartmouth's fiscal year.

A. SENIOR PERSONNEL

Juliette Bruce, Principal Investigator, (2.00 summer month per year) will oversee all aspects of the project and perform research as described in the project narrative. This proposed level of commitment is appropriate to the scope of work and in order to fulfill the research objectives.

The personnel funds requested are based on the individual annual salary multiplied by proposed effort.

B. OTHER PERSONNEL

TBD, Graduate Student, (12.00 calendar months per year) will perform research in topics related to the projects included in this proposal. In particular, developing computer software that helps compute various combinatorial invariants related to this project.

TBD, Undergraduate Students, will work on data generation, collection, and curation and website maintenance for a database connected to research in topics related to the projects included in this proposal. We estimate the undergraduate student will spend a total of ~100 hours per year working on the project.

C. FRINGE BENEFITS

The fringe benefit rate on salary support for all personnel (excluding the Graduate Student and Undergraduate Students) is budgeted at 34.5%. The fringe rate on the Undergraduate Student costs is 9% for all project years. There are no fringe benefit costs for the Graduate Student.

E. TRAVEL

1. Domestic: Travel funds of \$4,000 per year are requested to support 1 team member to attend a domestic conference for 6 days/5 nights and present results from this project. Possible conferences include Western Algebraic Geometry Seminar, the Joint Mathematics Meetings, and Sectionals of the American Mathematical Society. Costs follow standard college travel policies and are estimated as follows:

<u>Cost</u>	<u>Details</u>	<u>Qty</u>	<u>Rate</u>	<u>Per Person</u>	<u>Per Year</u>
Airfare	Flight to/from BOS	1	\$700	\$700	\$700
Conference Registration	Per Registration	1	\$1,000	\$1,000	\$1,000
Lodging	Per Night	5	\$320	\$1,600	\$1,600
Per Diem Meals/Incidentals	Per Day as per GSA	6	\$92	\$552	\$552
Ground transportation	Bus, parking, taxi, etc	1	\$148	<u>\$148</u>	<u>\$148</u>

Total Travel \$4,000 \$4,000

Travel funds are also requested for 1 team member to meet with collaborators for 4 days/3 nights. Possible meeting locations include U. of Minnesota, U. of San Diego, UC Berkeley, U. of Wisconsin, Texas A&M, Washington U., St. Louis, Cornell, and San Francisco State. Costs follow standard college travel policies and are estimated as follows:

<u>Cost</u>	<u>Details</u>	<u>Qty</u>	<u>Rate</u>	<u>Per Person</u>	<u>Per Year</u>
Airfare	Flight to/from BOS	1	\$700	\$700	\$700
Lodging	Per Night	3	\$320	\$960	\$960
Per Diem Meals/Incidentals	Per Day as per GSA	4	\$92	\$368	\$368
Ground transportation	Bus, parking, taxi, etc	1	\$148	<u>\$148</u>	<u>\$148</u>
		Total Travel		\$2,176	\$2,176

2. Foreign: Travel funds of \$5,000 per year are requested for 1 team member to attend an international conference for 6 days/5 nights and present results from this project. Possible conferences include Oberwolfach, MEGA, and the SIAM Applied Algebra Conference. Costs follow standard college travel policies and are estimated as follows:

<u>Cost</u>	<u>Details</u>	<u>Qty</u>	<u>Rate</u>	<u>Per Person</u>	<u>Per Year</u>
Airfare	Flight to/from BOS	1	\$900	\$900	\$900
Conference Registration	Per Registration	1	\$975	\$975	\$975
Lodging	Per Night	5	\$375	\$1,875	\$1,875
Per Diem Meals/Incidentals	Per Day as per GSA	6	\$184	\$1,104	\$1,104
Ground transportation	Bus, parking, taxi, etc	1	\$146	<u>\$146</u>	<u>\$146</u>
		Total Travel		\$5,000	\$5,000

G. OTHER DIRECT COSTS

4. Computer Services: Funds of \$5,000 are requested in year 1 for UM hosting of high-speed multi-parallel computations run on a computer maintained by the PI. Costs are estimated based on the current market prices of high-performance desktop computers/servers and the PI's experience.

5. OTHER:

- a. Health Insurance & Health Access Fees: Funds are requested in support of the Graduate Student at \$4,746 for health insurance and \$536 for fees in year 1 and increase by 5% annually.

I. INDIRECT COSTS

Project Year	Yr1	Yr2
MTDC Base	\$109,382	\$109,042
Total Indirect Costs Requested	\$69,458	\$69,700

In accordance with Dartmouth's negotiated F&A Rate Agreement with the Department of Health and Human Services (DHHS) dated 04/17/2024, the predetermined on-campus research rate is 63.5% from 7/1/2025 to 6/30/2027 and 64% from 7/1/2027 onward, calculated on a modified total direct cost (MTDC) base (MTDC = total direct costs excluding equipment, capital expenditures, charges for patient care, rental costs, tuition remission, scholarships and fellowships, participant support costs and the portion of each subaward in excess of \$25,000). The cognizant federal agency is DHHS, Darryl W. Mayes, Director, Division of Cost Allocation, 212-264-2069.

Facilities, Equipment, and Other Resources

Laboratory:

N/A

Clinical:

N/A

Animal:

N/A

Computer:

Dartmouth College and the Department of Mathematics at Dartmouth College have made significant investments in having the infrastructure necessary for large scale computation. Dartmouth Research computing has , “A total of 6000+ CPU cores, 120,000 GPU cores, 12+TB of memory, and 1 PB of storage across all systems” including many high-performance computing nodes. The Department of Mathematics has a number of its own servers dedicated to computational research. Dartmouth College also provides software licenses and access to Magma, Mathematica, Maple, R, Python, and other applications.

Space:

All faculty, graduate students, and visitors are provided with sufficient office space to conduct their teaching and research.

IDENTIFYING INFORMATION:

NAME: Bruce, Juliette

ORCID iD: <https://orcid.org/0000-0001-9766-7896>

POSITION TITLE: Assistant Professor

PRIMARY ORGANIZATION AND LOCATION: Dartmouth College, Mathematics, Hanover, New Hampshire, United States**Professional Preparation:**

ORGANIZATION AND LOCATION	DEGREE (if applicable)	RECEIPT DATE	FIELD OF STUDY
Brown University, Providence, Rhode Island, United States	Postdoctoral Fellow	08/2022 - 06/2024	Mathematics
University of California, Berkeley, Berkeley, California, United States	Postdoctoral Fellow	08/2020 - 07/2022	Mathematics
Mathematical Sciences Research Institute, Berkeley, California, United States	Postdoctoral Fellow	08/2020 - 05/2021	Mathematics
University of Wisconsin–Madison, Madison, Wisconsin, US	Ph.D	08/2020	Department of Mathematics
University of Wisconsin–Madison, Madison, Wisconsin, US	Masters of Arts	05/2016	Department of Mathematics
University of Michigan–Ann Arbor, Ann Arbor, Michigan, US	Bachelors of Science	04/2014	Department of Mathematics / Department of Political Science

Appointments and Positions

2024 - present Assistant Professor, Dartmouth College, Mathematics, Hanover, New Hampshire, United States

2025 - 2025 Long-Term Visiting Member, Fields Institute, Toronto, Ontario, Canada

2014 - 2020 Teaching Assistant / Research Assistant, University of Wisconsin - Madison, Department of Mathematics, Madison, WI, US

Products**Products Most Closely Related to the Proposed Project**

1. Madeline Brandt, Juliette Bruce, Melody Chan, Margarida Melo, Gwyneth Moreland, Corey Wolfe. On the top-weight rational cohomology of $\mathcal{A}_g$. Geometry & Topology. 2024 March. DOI: 10.48550/ARXIV.2012.02892
2. Bruce J, Erman D. A probabilistic approach to systems of parameters and Noether

normalization. Algebra & Number Theory. 2019 December 7; 13(9):2081-2102. Available from: <https://msp.org/ant/2019/13-9/p05.xhtml> DOI: 10.2140/ant.2019.13.2081

3. Juliette Bruce. Asymptotic syzygies for products of projective spaces. Journal of Algebra. 2024 July. DOI: 10.1016/j.jalgebra.2024.03.013
4. Juliette Bruce, Lauren Cranton Heller, Mahrud Sayrafi. Characterizing Multigraded Regularity on Products of Projective Spaces. 2021. Available from: <https://arxiv.org/abs/2110.10705> DOI: 10.48550/ARXIV.2110.10705
5. Juliette Bruce, Lauren Cranton Heller, Mahrud Sayrafi. Bounds on Multigraded Regularity. 2022. Available from: <https://arxiv.org/abs/2208.11115> DOI: 10.48550/ARXIV.2208.11115

Other Significant Products, Whether or Not Related to the Proposed Project

1. Bruce J, Erman D, Goldstein S, Yang J. Conjectures and Computations about Veronese Syzygies. Experimental Mathematics. 2018 June 21; 29(4):398-413. Available from: <https://www.tandfonline.com/doi/full/10.1080/10586458.2018.1474506> DOI: 10.1080/10586458.2018.1474506
2. Juliette Bruce. The quantitative behavior of asymptotic syzygies for Hirzebruch surfaces. Journal of Commutative Algebra. 2022 March. DOI: 10.1216/jca.2022.14.19
3. Bruce J, Corey D, Erman D, Goldstein S, Laudone R, Yang J. Syzygies of $\mathbb{P}^1 \times \mathbb{P}^1$: Data and conjectures. Journal of Algebra. 2022 March; 593:589-621. Available from: <https://linkinghub.elsevier.com/retrieve/pii/S0021869321005160> DOI: 10.1016/j.jalgebra.2021.10.023
4. Ayah Almousa, Juliette Bruce, Michael Loper, Mahrud Sayrafi. The virtual resolutions package for Macaulay2. Journal of Software for Algebra and Geometry. 2020 August. DOI: 10.2140/jsag.2020.10.51
5. Eran Assaf, Madeline Brandt, Juliette Bruce, Melody Chan, Raluca Vlad. Tropicalizations of locally symmetric varieties. 2025. Available from: <https://arxiv.org/abs/2505.10504> uri: <https://arxiv.org/abs/2505.10504>

Certification:

I certify that the information provided is current, accurate, and complete. This includes but is not limited to information related to domestic and foreign appointments and positions.

I also certify that, at the time of submission, I am not a party to a malign foreign talent recruitment program.

Misrepresentations and/or omissions may be subject to prosecution and liability pursuant to, but not limited to, 18 U.S.C. §§ 287, 1001, 1031 and 31 U.S.C. §§ 3729-3733 and 3802.

Certified by Bruce, Juliette in SciENCv on 2025-09-13 12:01:47

Other Personnel Biographical Information

Data Not Available

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Bruce, Juliette

PERSISTENT IDENTIFIER (PID) OF THE SENIOR/KEY PERSON: <https://orcid.org/0000-0001-9766-7896>

*POSITION TITLE: Assistant Professor

*ORGANIZATION AND LOCATION: Trustees of Dartmouth College, Hanover, New Hampshire, United States

Proposals/Active Projects

*Proposal/Active Project Title: Multigraded Homological Algebra & Geometry

*Status of Support: Pending

Proposal/Award Number: 277777

*Source of Support: NSF

*Primary Place of Performance: Dartmouth College, Hanover NH USA

*Proposal/Active Project Start Date: (MM/YYYY): 05/2026

*Proposal/Active Project End Date: (MM/YYYY): 04/2028

*Total Anticipated Proposal/Project Amount: \$357,582

* Person Months per budget period Devoted to the Proposal/Active Project:

Year	Person Months
2027	2
2028	2

*Overall Objectives: This proposal develops new connections between multigraded homological algebra and geometry. The first project expands our understanding of the relationship between minimal graded free resolutions and geometry by studying the syzygies of stacky curves and stacks embedded by log weighted series.. The second project is devoted to understanding multigraded regularity on toric varieties with connections to binational geometry and combinatorial algebraic geometry.

*Statement of Potential Overlap: None

Certification:

I certify that the information provided is current, accurate, and complete. This includes but is not limited to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

I also certify that, at the time of submission, I am not a party to a malign foreign talent recruitment program.

Misrepresentations and/or omissions may be subject to prosecution and liability pursuant to, but not limited to, 18 U.S.C. §§ 287, 1001, 1031 and 31 U.S.C. §§ 3729-3733 and 3802.

Certified by Bruce, Juliette in SciENcv on 2025-09-15 13:44:48

Table 1

1	Your Name:	Your Organizational Affiliation(s), last 12 mo	Last Active Date
	Bruce, Juliette E	Dartmouth College	

Table 2

2	Name:	Type of Relationship	Optional (email, Department)	Last Active Date
R	Newton, Kit	Family	knewton@dvc.edu	

Table 3

3	Advisor/Advisee Name:	Organizational Affiliation	Optional (email, Department)
G	Ermann, Daniel	University of Hawaii	erman@math.hawaii.edu
T	DeGroot, William	Dartmouth College	william.h.degroot.gr@dartmouth.edu

Table 4

4	Name:	Organizational Affiliation	Optional (email, Department)	Last Active Date
C	Ardila, Federico	San Francisco State University	federico@sfsu.edu	06/01/24
A	Assaf, Eran	Massachusetts Institute of Technology	eranasaf@mit.edu	
C	Banks, Maya	University of Illinois, Chicago	mayadb@uic.edu	
C	Beck, Matthias	San Francisco State University	mattbeck@sfsu.edu	06/01/24
C	Berkesch, Christine	University of Minnesota - Twin Cities	cberkesch@umn.edu	
A	Bonato, Anthony	Ryerson University	abonato@ryerson.ca	01/01/24
C	Boocher, Adam	University of San Diego	aboocher@sandiego.edu	06/01/24
A	Brandt, Madeline	Vanderbilt University	madeline.v.brandt@vanderbilt.edu	
C	Bucciarelli, Jacob	Kansas State University	jbucciarelli@ksu.edu	
C	Buckmire, Ron	Occidental College	ron@oxy.edu	01/01/24
A	Cavalieri, Renzo	Colorado State University	renzo@math.colostate.edu	
A	Chan, Melody	Brown University	melody_chan@brown.edu	
A	Cobb, John	Auburn University	jdcobb3@gmail.com	
A	Corey, Daniel	University of Nevada - Las Vegas	daniel.corey@unlv.edu	
A	Cranton Heller, Lauren	University of Nebraska - Lincoln	lheller2@unl.edu	
A	Eisenbud, David	University of California - Berkeley	de@math.berkeley.edu	07/31/22
C	Ermann, Daniel	University of Hawaii	erman@math.hawaii.edu	
C	Hering, Milena	Edinburgh University	m.hering@ed.ac.uk	
C	Hosten, Serkan	San Francisco State University	serkan@sfsu.edu	06/01/24
C	Huang, Daoji	Institute for Advanced Study	dhuang@ias.edu	

C	Kelly, Tyler, L	University of Birmingham	t.kelly.1@bham.ac.uk	08/01/22
C	Klein, Patricia	Texas A&M University	pjklein@tamu.edu	06/01/25
A	Maclagan, Diane	University of Warwick	D.Maclagan@warwick.ac.uk	
A	Manon, Christopher	University of Kentucky	christopher.manon@uky.edu	
C	Maclagan, Diane	University of Warwick	D.Maclagan@warwick.ac.uk	
A	Melo, Margarida	Roma Tre University	melo@mat.uniroma3.it	07/01/22
A	Moreland, Gwyneth	Harvard University	gwynm@math.harvard.edu	07/01/22
C	Ramkumar, Ritvik	University of Notre Dame	rramkuma@nd.edu	07/31/22
C	Ross, Dustin	San Francisco State University	rossd@sfsu.edu	
A	Sayrafi, Mahrud	McMaster University	mahrud@mcmaster.ca	
C	Seceleanu, Alexandra	University of Nebraska - Lincoln	aseceleanu@unl.edu	
A	Sayrafi, Mahrud	University of Tulane	cwolfe@tulane.edu	07/01/22
A	Vlad, Raluca	Brown University	raluca_vlad@brown.edu	
C	Voight, John	Dartmouth College	jvoight@gmail.com	08/01/22

Table 5

5	Name:	Organizational Affiliation	Journal/Collection	Last Active Date
B	Huybrechts, Daniel	Universität Bonn	Journal für die reine und angewandte Mathematik	
B	Payne, Sam	University of Michigan	Selecta Mathematica	07/01/25

SYNERGISTIC ACTIVITIES

- **Research Conference Organizing:** The PI has organized many national research conferences and workshops. Most recently the PI was a co-organizer for the Fall 2024, *Algebraic Geometry Northeastern Series (AGNES)*, a weekend conference with 100 participants held at Dartmouth College aimed at introducing early career researchers (e.g. graduate students and postdocs) to current research trends in algebraic geometry and related fields.
- **Developed Open-Source Software:** The PI has a long commitment to developing open-source software to promote and support research in algebraic geometry and commutative algebra. For example, the PI developed the *VirtualResolutions* package for the computer algebra system *Macaulay2*. This package, which includes core functionality for multigraded homological algebra on toric varieties, has been proven beneficial to many research projects and is frequently cited.
- **Invited Research Talks:** The PI has given many invited research talks including giving a plenary talk *Homological Algebra Beyond Projective Space* at $\text{Spec}(\overline{\mathbb{Q}}(2\pi i))$ a research conference held at the Fields Institute.
- **Preparing Future Researchers** The PI has organized and lead numerous initiatives and conferences aimed at helping graduate students and early career researchers establish themselves within the field, find research mentors, and a community of collaborators. Of note is the *Gems Of...* series of weekend and week-long workshops that the PI organized, with NSF support, for young researchers in commutative algebra and combinatorics. This series has reached over 150+ participants over 3 years.
- **Growing the Mathematics Community** The PI served as the inaugural president for *Spectra*, aimed at widening the mathematical community. This included growing the organization's budget by \$20,000 and its membership to over 500 people, and launching the organization's annual invited lecture series at the Joint Mathematics Meeting.

Data Management Plan

The PI envisions the only data generated from this proposal to be

- (1) mathematical articles and pre-prints,
- (2) computer code written for use in my research, and
- (3) computational examples generated for my research.

All mathematical articles will be made publicly available via the arXiv pre-print server, before eventually being submitted to peer-reviewed journals. Any computer code written or examples generated, which might be interesting to others, will be made publicly accessible from the PI's website or GitHub repository.

MENTORING PLAN

Orientation: The PI will meet with all students and post-docs (none planned) associated with this grant to have a careful conversation surrounding the goals of this proposal, their career and educational goals, and the expectations for participating in these projections. These expectations and commitments from both the student/post-doc and the PI will be recorded in writing and frequently used as a benchmark to judge progress and make adjustments.

Professional Advising: In addition to the research advising the PI will provide students/post-docs with mentorship and advice on how they can move towards achieving their professional goals. This will include both informal advising and formal advising at scheduled monthly meetings specifically designated towards conversations around professional goals.

Preparing Publications: The PI will provide the students/postdocs with advice and suggestions for how to approach academic publishing, both in theory and in practice with any publications that may result from this work.

Networking: The PI will help students and post-docs identify and apply to attend conferences and workshops appropriate for building their professional and research networks.

Data Not Available

List of Suggested Reviewers

Data Not Available

List of Reviewers Not to Include

Data Not Available